BARCS: beta-binomial analysis and regression for quantitative and covariate-adjusted CRISPR pooled screens

Hyun-Hwan Jeong

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Abstract

BARCS was strongest when a pooled screen contained information beyond one binary contrast. In a longitudinal HT-29 screen, it achieved the highest recall at 90% precision while essentially matching continuous-time MAGeCK-MLE in global discrimination. In a primary-T-cell FACS screen, control-calibrated BARCS recovered 22 of 26 experimentally confirmed IL2RA regulators versus 17 for all-bin MAGeCK-MLE and reduced the non-targeting $p < 0.05$ frequency from 13.3% to 4.9%. In a 10,000-gene CRISPulator FACS simulation, dispersion-moderated BARCS matched the count models and MAGeCK-MLE in ranking quality (average precision 0.919 versus 0.916–0.921) while realizing a false-discovery proportion of 0.066 against a requested 0.10, where edgeR-QL, DESeq2, and limma-voom realized 0.21 to 0.24. Conversely, Liang's published RRA led the processed-count Cas13 benchmark, exposing the power boundary of regression inference with one residual degree of freedom.

The problem addressed is coefficient inference from pooled-screen counts when independent biological screen units, represented by libraries, lie on a quantitative or covariate-adjusted design. BARCS extends the depth-aware beta-binomial principle of CB^2 from two group means to $\mathrm{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g$ and tests a prespecified coefficient or contrast. Full-library totals determine sequencing precision, extra-binomial dispersion limits the information gained from depth, and residual degrees of freedom come from independent libraries. Fisher or Stouffer aggregation is not required by the guide-level model; an optional shared-effect Wald procedure provides exploratory gene inference when several guides target one gene. Parallel RcppArmadillo kernels analyzed 86,882 guides in 19.4 seconds. BARCS therefore solves a specific gap between pairwise testing and specialist generative models: transparent, scalable inference for quantitative and adjusted pooled-screen designs, with explicit diagnostics for designs it cannot repair.

1 Introduction

Many pooled CRISPR screens no longer ask only whether guide abundance differs between a control group and one treatment group. Libraries may be collected across time, dose, ordered FACS bins, donors, batches, or combinations of these variables. The scientific target is then a trend, adjusted association, interaction, or prespecified contrast. Dichotomizing the design into "high" and "low" groups discards intermediate information and makes covariate adjustment opaque.

The unmet statistical problem can be stated directly. For guide g, suppose we observe counts K_{gi}, full mapped-guide totals N_i, and one row $\mathbf{x}_i^\top$ of a full-rank design matrix for each independent biological screen unit i, represented by a sequenced library. We want to estimate

$$\mathrm{logit}(\mu_{gi}) = \mathbf{x}_i^\top \boldsymbol{\beta}_g \quad \text{and test} \quad H_0 : \mathbf{c}^\top \boldsymbol{\beta}_g = 0, \tag{1}$$

while allowing unequal library depths and extra-binomial variation. The output should be an interpretable guide coefficient, standard error, and p-value for the scientific contrast selected before examining the guide results.

Terminology matters: guide count or proportion is the response in Equation (1). Time, dose, treatment, or an ordered phenotype summary enters as a sample-level predictor. BARCS does not analyze a continuous per-cell outcome unless that outcome has first been observed or validly summarized at the library level.

Table 1: Operational definition of the problem BARCS addresses.

Component	Definition
Observed data	Guide counts K_{gi}, unfiltered full-library totals N_i, and a prespecified sample design matrix X
Primary estimand	A guide-level coefficient or contrast $\mathbf{c}^\top \boldsymbol{\beta}_g$ conditional on the other columns of X
Information unit	An independent biological screen unit represented by a sequenced library; additional reads refine that unit but do not create another one
Primary output	Guide effect, standard error, t statistic, residual degrees of freedom, p-value, FDR, and dispersion diagnostics
Optional gene output	An inverse-variance shared-effect Wald statistic with empirical-null calibration; no Fisher or Stouffer step is required
Not identified	Unobserved single-cell phenotypes, dependence among bins from one biological pool, mechanistic growth dynamics, or uncertainty that would have required missing biological replicates

This is not solved by simply applying ordinary logistic regression to every read. Reads measure one library more precisely but are not independent biological replicates. Nor is it solved by running a two-group method repeatedly: separate pairwise tests cannot represent one adjusted coefficient and needlessly fragment the available design. At the other extreme, specialist models such as Chronos or Waterbear answer richer mechanistic or joint-bin questions but are not general-purpose coefficient interfaces.

BARCS—Beta-binomial Analysis and Regression for CRISPR Screens—fills this gap. It models each guide count against the corresponding full-library total, uses a beta-binomial variance to separate sequencing precision from between-library heterogeneity, and places continuous variables, factors, interactions, and covariates in an ordinary R model matrix. Independent libraries, not read depth or guide number, determine the sample-level reference degrees of freedom.

The manuscript makes four contributions. First, it defines the quantitative pooled-screen target as a coefficient or contrast rather than a forced binary comparison. Second, it establishes precisely how that regression framework relates to the original CB^2 statistic: exact nesting on the legacy group-summary scale and first-order local equivalence for the default logit fit under stated asymptotics, not finite-sample identity. Third, it provides control-based diagnostics and an optional shared-effect gene statistic without making Fisher or Stouffer aggregation part of the core method. Fourth, it evaluates both the intended use case and failure boundaries against MAGeCK, Chronos, BAGEL2, Waterbear, MAUDE, and published rank aggregation.

The scope is deliberately narrower than continuous per-cell phenotype analysis. BARCS analyzes a quantitative coordinate attached to each biological screen unit. It cannot reconstruct a continuous phenotype for each cell when guide identity and phenotype were never observed jointly. It also does not make correlated FACS bins independent, infer mechanistic viability dynamics, correct copy number automatically, or replace missing biological replication. Those boundaries determine the benchmark design and the method choice rules reported below.

2 Results

The evaluation asks four questions rather than seeking one universal winner: does the full quantitative design recover biology that a pairwise split discards; are thresholded discoveries calibrated; what happens when replication or the likelihood interpretation is inadequate; and does the method remain competitive in ordinary endpoint screens? We therefore begin with a longitudinal intended use case, examine a low-replication processed-count boundary, stress-test ordered-bin FACS data, and then use controlled simulations, backward-compatibility screens, and copy-number diagnostics to delimit the claim. Method colors are fixed throughout: BARCS is blue, MAGeCK orange, Chronos green, published endpoint MAGeCK pink, BAGEL2 gold, Waterbear purple, and MAUDE red.

2.1 Primary use case: longitudinal HT-29 screening

The central use case for BARCS is a continuous predictor observed at more than two values. We therefore begin with the HT-29 time course of Tzelepis et al. [1], the longitudinal dataset used to evaluate Chronos [2]. It contains a pDNA measurement and HT-29 guide counts at days 7, 10, 13, 16, 19, 22, and 25. The three sequencing columns at each late day were summed before analysis. This distinction is important for the requested t inference: three assays of one harvested population increase read depth but are not three independent biological units and therefore must not add residual degrees of freedom. Following the Chronos preprocessing, guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes.

The BARCS and official MAGeCK-MLE 0.5.9.5 fits received the same eight-column count matrix and the same numeric design, with an intercept and $d_i/25$. This is a direct continuous-time comparison. We also downloaded the all-seven-time-point Chronos joint result and the deposited per-endpoint MAGeCK and BAGEL2 results from the Chronos Figshare record [3]. As in the Chronos paper, the latter two were combined by taking each gene's median effect over the seven separate endpoint runs.

For an external biological target, core essential genes are positives and HT-29 genes with DepMap 20Q2 expression below 0.5 are negatives, matching the deposited Chronos analysis. All methods were restricted to the same 1,080 essential and 5,774 unexpressed genes. Normalized null-median difference (NNMD) is the essential-gene median minus the unexpressed-gene mean, divided by mean absolute deviation of the unexpressed effects; more negative values indicate stronger separation. "False positives" counts unexpressed genes in the lowest 15% of all evaluated effects, following the deposited Chronos analysis.

Table 2: Longitudinal HT-29 control-gene benchmark. PR AUC is the area under the precision–recall curve; Recall_{90} is the maximum recall at least 90% precision. Higher AUROC, PR AUC, and Recall_{90} are better; more negative NNMD and fewer unexpressed false positives are better. The best value in each column is shown in bold and in its method color; row swatches use the same colors as Figure 1.

Method	AUROC	PR AUC	NNMD	Recall_{90}	$\mathrm{FP}_{15\%}$
■ BARCS time	0.9786	0.9494	−12.35	**0.9037**	**74**
■ Official MAGeCK time	**0.9789**	**0.9534**	−13.14	0.8972	76
■ Chronos joint	0.9747	0.9356	**−18.88**	0.8991	77
■ Published MAGeCK average	0.9742	0.9351	−10.62	0.8602	100
■ Published BAGEL2 average	0.9725	0.9295	−7.28	0.8148	133

Table 2 gives a more useful answer than declaring one universal winner. Official continuous-time MAGeCK has the highest AUROC and PR AUC, although its AUROC advantage over BARCS is only 0.0004. BARCS has the highest high-precision recall and two fewer unexpressed false positives. Chronos has by far the strongest distributional separation by NNMD, consistent with its mechanistic treatment of knockout efficiency and saturation. The BARCS and official continuous-time MAGeCK effects have Spearman correlation 0.926; correlations with Chronos joint and the deposited MAGeCK average are 0.815 and 0.812. These are again effect agreement statistics, not comparable likelihoods: the methods estimate differently scaled quantities and do not expose a common predictive likelihood in the deposited outputs.

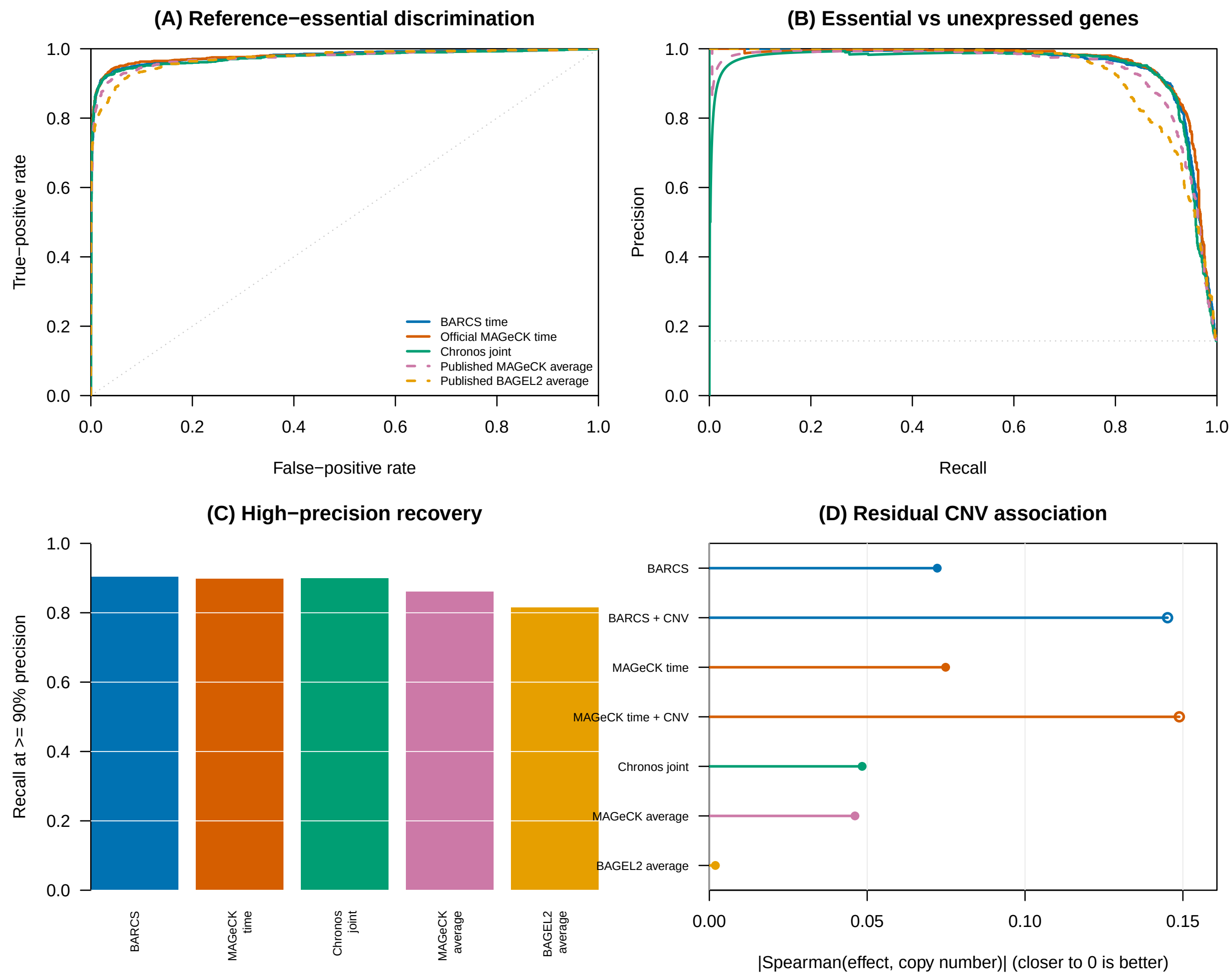

Figure 1: Longitudinal HT-29 benchmark. (A) ROC and (B) precision–recall curves compare essential-gene recovery against unexpressed genes. (C) Recall at at least 90% precision. (D) Absolute residual effect–copy-number association among unexpressed genes; values closer to zero are better. Filled points are uncorrected results and open points are results after official MAGeCK piecewise correction. A method has the same color in every panel.

2.1.1 What the longitudinal screen says about CNV correction

We extracted the HT-29 profile (ACH-000552) from DepMap Public 20Q2 [4] and repeated the same copy-number audit used for A375. Before correction, effect–copy-number Spearman correlation among unexpressed genes is already small: −0.072 for BARCS, −0.075 for official continuous-time MAGeCK, −0.048 for deposited Chronos joint effects, −0.046 for the published MAGeCK average, and −0.002 for the published BAGEL2 average. The near-zero BAGEL2 association is a focused robustness diagnostic, not evidence that BAGEL2 is the best classifier: Table 2 shows that it has the lowest discrimination and high-precision recovery on the same control-gene universe. Applying MAGeCK 0.5.9.5's official piecewise normalizer makes the association larger in absolute value rather than smaller: −0.145 for BARCS and −0.149 for MAGeCK. Thus the A375 result does not generalize automatically.

This failure mode is informative. The piecewise correction estimates one global effect–CNV trend across genes in one cell line and applies it after model fitting. When the nuisance association is weak and biological dependencies are not exchangeable across copy-number regions, the fitted trend can have the wrong practical target and over-correct null genes. Chronos joint is shown without CNV correction because the published Chronos procedure requires multiple cell lines to identify common essential genes and was explicitly not applied to this single-line experiment [2]. The DepMap profile was measured separately from the original screen, so cell-line drift is an additional limitation; the result supports diagnosing correction rather than applying it by default.

2.2 Processed-count boundary: Liang Cas13 fitness screens

The five-cell-line Cas13 fitness study of Liang et al. [5] provides a useful head-to-head boundary because it contains a specialist published analysis, known-essential protein-coding controls, and cell-line-specific non-expressed lncRNA null controls. We used the deposited normalized Table S2 values requested for this comparison. They are fractional after median-of-ratios normalization, ComBat correction, and replicate-outlier processing. We rounded them once to the nearest pseudo-count and supplied the identical matrix, with no additional normalization, to BARCS, official MAGeCK-RRA, and official MAGeCK-MLE. Liang RRA denotes the authors' deposited `RobustRankAggreg` result and is not MAGeCK-RRA.

Table 3: Liang Cas13 processed-count sensitivity analysis, macro-averaged over five cell lines. $\text{Recall}_{5\%}$ fixes the empirical false-positive rate among non-expressed lncRNA null controls at 5%. Calibration error is the absolute difference between the observed null $p < 0.05$ fraction and 0.05, so lower is better. The other four metrics are higher-is-better. The best value in each column is shown in bold and in its method color.

Method	AUROC	Avg. precision	$\text{Recall}_{5\%}$	Calibration error	$\text{FDR}_{0.10}$ recall
■ Liang RRA	**0.9703**	**0.8908**	**0.9167**	**0.0174**	**0.8233**
■ Official MAGeCK-RRA	0.9522	0.8425	0.8733	0.0181	0.7400
■ Official MAGeCK-MLE	0.9508	0.8356	0.8600	0.0710	0.6800
■ BARCS	0.9386	0.7756	0.8233	0.0274	0.4867

Liang RRA leads every macro-averaged metric in Table 3; BARCS does not beat it. At a common empirical 5% null false-positive rate, Liang RRA recovers 91.7% of essential controls versus 82.3% for BARCS. At the 0.10 gene-FDR threshold, the gap widens to 82.3% versus 48.7%. The latter difference is consistent with finite-sample conservatism: every cell line supplies only three or four libraries, and the fitted day effect has one residual degree of freedom after the intercept and available replicate block. K562 is the sharpest example: BARCS retains AUROC 0.911 but calls none of the essential controls at FDR 0.10. More sequencing depth cannot replace the missing independent libraries in a sample-level t reference.

Official MAGeCK-RRA also modestly outperforms MAGeCK-MLE here. Its null calibration error is 0.018 compared with 0.071 for MLE, while its average precision is 0.843 compared with 0.836. This pattern supports a design-specific interpretation: guide-rank aggregation is well matched to a two-endpoint study with few replicates and already normalized inputs, whereas a regression likelihood has little sample-level information from which to estimate uncertainty. It is not evidence that RRA is universally superior to count regression.

Effect agreement remains substantial despite the threshold differences. Within each cell line we computed Spearman correlation across all genes with finite effects, then averaged the five correlations. BARCS correlates 0.875 with MAGeCK-MLE, 0.864 with MAGeCK-RRA, and 0.784 with Liang RRA. Thus the principal BARCS deficit in this benchmark is not a reversal of the biological ranking; it is reduced resolution and power under one residual degree of freedom.

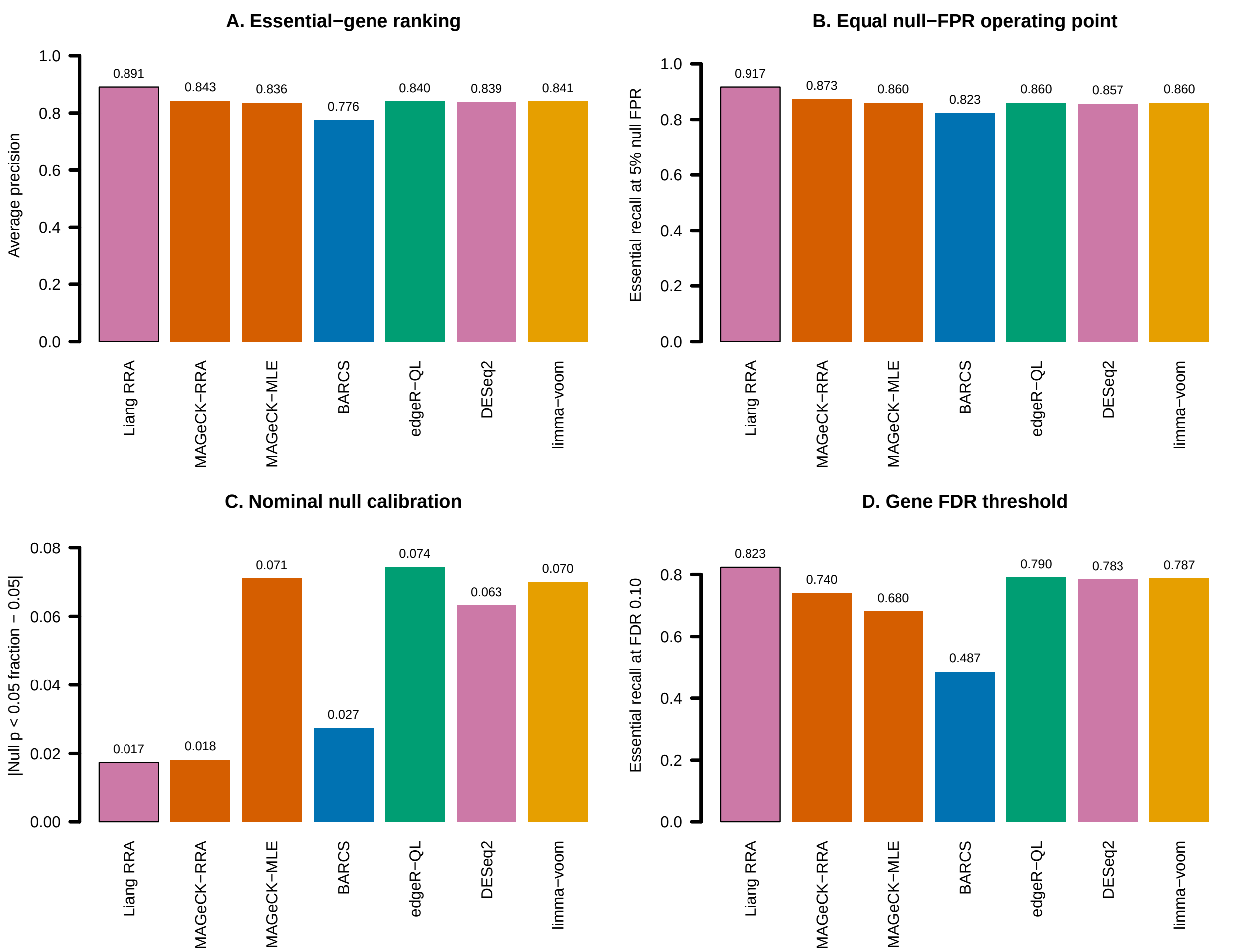

Figure 2: Liang Cas13 processed-count sensitivity analysis, macro-averaged over five cell lines. (A) Average precision for ranking essential controls above non-expressed lncRNA nulls. (B) Essential-control recall at a common empirical 5% null false-positive rate. (C) Absolute error from the nominal 5% null p-value rate; lower is better. (D) Directional essential-control recall at gene FDR 0.10. A black outline marks the best method in each panel.

This comparison has an important limit. Rounding changes each deposited value by only 0.25 on average and by less than 0.5 at most, but the values were transformed before deposition. Consequently, neither the beta-binomial nor negative-binomial likelihood has its literal raw-count sampling interpretation, and the nominal calibration panels are sensitivity diagnostics rather than raw-count likelihood validation. The reproducible streaming pipeline supplied with BARCS makes a future raw-read confirmation possible; until that is complete, the defensible conclusion is that the published Liang RRA is the best fit to the deposited processed data.

2.3 Stress test and model boundary: ordered-bin FACS

The Waterbear study supplies a complementary continuous-phenotype benchmark with unusually useful validation [6]. GSE242880 contains a pooled CRISPR screen of IL2RA protein abundance in primary human T cells. Its low-coverage, high-MOI arm has four ordered FACS bins for each of three donors, 6,000 guides targeting 1,350 genes or negative controls, and 26 regulators whose direction was confirmed by individual knockout and flow cytometry. Because these are primary donor cells rather than a cancer cell line, this benchmark is not driven by tumor copy-number artifacts.

We assigned the four bins latent marker locations equal to the conditional means of standard-normal intervals with the prespecified bin masses $(0.2, 0.3, 0.3, 0.2)$. Beta-binomial regression used all 12 libraries with a marker slope and donor indicators. Official MAGeCK-MLE received the same model space; its marker score was affinely mapped to zero–one because the MAGeCK initializer requires a reference row with all non-intercept entries zero. We also fitted two outer-bin ablations and ran the paper's native `mageck test` comparison of Q1 with Q4.

This design exposes a limitation of applying independent-library regression to sorted data. The four bins from one donor partition one cell pool and are therefore negatively correlated, while the beta-binomial fit treats their margins separately. Among 593 non-targeting guides, the raw all-bin beta-binomial $p < 0.05$ frequency is 0.133. We therefore added the prespecified negative-control calibration implemented by `bb_calibrate_controls()`: the empirical 95th percentile of absolute control t statistics is matched to the t_8 two-sided 0.05 cutoff, with scales below one prohibited. The estimated scale is 1.357; after calibration, the non-targeting frequency is 0.049. Estimates and the t reference are retained, while raw and calibrated inferential columns are both saved.

Table 4: GSE242880 low-coverage, high-MOI IL2RA FACS benchmark. Recovery uses each method's reported 0.10 discovery threshold and the independently validated direction. BARCS and MAGeCK use adjusted p-values; Waterbear uses posterior inclusion probability above 0.90. The non-targeting column is guide-level. Waterbear and MAUDE values are reported by the study and were not rerun. The highest validated recovery and the available non-targeting rate closest to 0.05 are highlighted.

Method	Discoveries	Validated recovery	NTC $p < 0.05$
■ BARCS, four bins + controls	49	22/26	**0.049**
■ BARCS, four bins raw	127	23/26	0.133
■ Official MAGeCK-MLE, four bins	72	17/26	—
■ BARCS, outer bins	34	18/26	0.064
■ Official MAGeCK test, outer bins	32	18/26	—
■ Waterbear, four bins (reported)	79	24/26	—
■ MAUDE, four bins (reported)	406	**25/26**	—

The calibrated four-bin BARCS result recovers four more validated regulators than all-bin MAGeCK-MLE while making 23 fewer calls. It is two regulators behind the reported Waterbear result and three behind MAUDE. The raw 23/26 BARCS result is not the primary claim because its negative controls reveal inflation. MAUDE's 25/26 recovery is also not evidence of superior specificity: it calls 406 of 1,350 genes, and the Waterbear study found poor MAUDE calibration in simulations [6, 7]. Across all 1,350 genes, uncalibrated BARCS and MAGeCK-MLE all-bin effects have Spearman correlation 0.917, so the recovery difference is not caused by globally incompatible directions. All 26 validation effects have the expected sign under each independently rerun method.

The follow-up table contains more information than the 26-positive subset. Thirty-three candidates were tested individually: 26 had a concordant IL2RA effect and seven did not validate. Restricting evaluation to these measured labels avoids the incorrect assumption that every other screen gene is a false positive. Table 5 reports threshold and ranking metrics for the five methods rerun here.

Table 5: Classification in the selected 33-gene experimental follow-up panel (26 concordant positives and seven non-validating candidates). This panel is small and was selected from a previous screen, so the metrics are supporting evidence rather than unbiased genome-wide performance estimates. Higher is better for every metric; column maxima are highlighted.

Method	F1	MCC	Balanced accuracy	AUROC	Avg. precision
■ BARCS, four bins + controls	**0.863**	**0.398**	**0.709**	0.808	0.945
■ BARCS, four bins raw	0.852	0.194	0.585	**0.819**	**0.950**
■ Official MAGeCK-MLE, four bins	0.723	0.070	0.541	0.626	0.872
■ BARCS, outer bins	0.783	0.340	0.703	0.714	0.913
■ Official MAGeCK test, outer bins	0.766	0.224	0.632	0.703	0.906

The raw four-bin model has the strongest rank metrics, which indicates that the continuous trend is extracting useful signal. Control calibration changes the decision boundary rather than the estimated effects and produces the best F1, Matthews correlation, and balanced accuracy in the validation panel. That combination is the central evidence for BARCS here: continuous-bin ranking gains are retained while explicit null guides prevent the extra calls from being mistaken for confidence.

This benchmark is stronger than a rank-correlation-only comparison because it contains directionally validated biology and hundreds of explicit null guides. It is not an unbiased precision experiment: the 26 genes were chosen for follow-up from an earlier screen, and unvalidated genes cannot simply be called false positives. Nor does it overturn Waterbear's modeling advantage. Waterbear represents the joint multinomial bin structure, uncertain cutoffs, guide-to-gene hierarchy, and sparse effects; BARCS supplies a much faster transparent trend test but needs negative controls here to diagnose and scale inference under a violated independence assumption. The deposited paper reports 24/26 for Waterbear and 17/26 for MAGeCK; the independent current `mageck test` rerun gives 18/26, a one-gene implementation-level difference.

These results separate three meanings of "better." First, ranking quality asks whether validated regulators move toward the top; the raw four-bin model performs strongly by AUROC and average precision. Second, decision quality asks whether a prespecified threshold balances recovery and false calls; the control-calibrated model leads the rerun methods by F1 and Matthews correlation. Third, likelihood fidelity asks whether the sampling dependence is represented; Waterbear wins that design question because it models all bins from a donor jointly. BARCS is compelling here because it does well on the first two questions with a simple general-purpose model and also provides the control diagnostic that exposes the third.

The published Waterbear and MAUDE aggregates cannot be assigned validation- panel F1 or Matthews correlation without their gene-level calls for all seven non-validating candidates. Reporting

24/26 or 25/26 as if it were complete accuracy would therefore mix sensitivity with an unknown precision. We retain those values as positive-control recovery references and restrict classification metrics to the methods rerun from complete outputs.

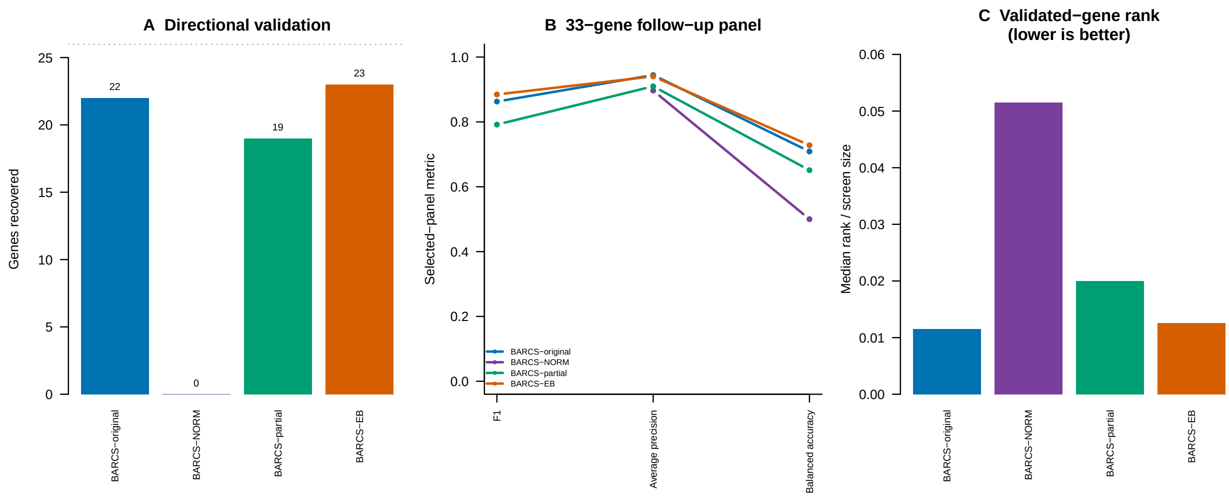

Figure 3: Ordered-bin IL2RA FACS benchmark. (A) Positive-control recovery for all methods. (B) F1 in the selected 33-gene panel for methods rerun here. (C) All-bin BARCS and MAGeCK-MLE effects, with validated genes highlighted. (D) Non-targeting-guide empirical null curves, showing the correction of raw all-bin inflation.

2.4 Genome-scale FACS simulation: does dispersion moderation change the calibration–power tradeoff?

To isolate the FACS design and estimation questions from the candidate-selected Waterbear validation set, we used CRISPulator 0.5.1 [8] to simulate CRISPR knockout screens with known gene effects at genome scale. Each of three prespecified seeds generated 10,000 genes, five guides per gene, and four independent screen replicates, for 50,000 guides per screen. Twenty percent of genes changed the latent phenotype and 5% were explicit negative controls. The stress-test setting used phenotype noise $\sigma = 2$, 50 sorted cells per guide, and 50 reads per guide per sample. Multiplicity of infection was 0.20 in the main analysis and 0.30 in the supplementary analysis: 0.20 is the operating point at which the single-integration approximation shared by all of these guide-level models is most defensible, and 0.30 stresses it.

Each replicate produced a low 0–25% tail, a high 75–100% tail, and an unsorted 0–100% bulk sample. The two tails were assigned the conditional standard-normal locations -1.271 and 1.271; bulk was assigned zero. Replicate indicators were included in every fit. We compared this three-sample fit with the two-tail ablation. Importantly, "three sample" does not mean three independent bins: the bulk sample contains the cell populations from both tails and costs 50% more sequencing than the tail-only design.

Two BARCS gene statistics are carried here. BARCS-ST (`BARCS-original`) is the historical calibrated signed-z statistic. BARCS-MOD (`BARCS-moderated`) reuses the same guide fits after guide-dispersion moderation, which shrinks each guide's variance inflation toward a library-wide trend before the same signed-z aggregation. At genome scale the moderation prior is estimated from 50,000 guides and is correspondingly sharp: the fitted prior degrees of freedom were 42.7–45.8 against the seven residual degrees of freedom the design itself supplies.

Table 6: Genome-scale CRISPulator FACS benchmark at MOI 0.20, 10,000 genes and 50,000 guides, over three fixed simulation seeds (mean ± standard deviation). All methods are scored on the genes every method returned a finite result for (9,475–9,517 per run); MAGeCK-MLE omits negative-control genes from its gene summary once their sgRNAs are declared, so the common set excludes them. Recall counts active genes called at gene FDR 0.10 with the correct direction. FDP is the realized fraction of inactive genes among calls; the nominal target is 0.10. Column maxima for ranking, effect recovery, recall, and F1 are highlighted.

Method	Samples	AUROC	Avg. precision	Effect ρ_s	Directional recall	FDP	F1
■ BARCS-ST	Low–bulk–high	0.947±.004	0.902±.007	0.924±.004	0.716±.022	0.065±.006	0.811±.012
■ BARCS-MOD	Low–bulk–high	0.954±.003	0.919±.005	0.924±.004	0.774±.014	0.066±.002	**0.847±.008**
■ MAGeCK-MLE	Low–bulk–high	**0.957±.002**	**0.921±.003**	**0.933±.002**	0.563±.030	0.006±.002	0.718±.024
■ edgeR-QL + Stouffer	Low–bulk–high	0.953±.003	0.916±.005	0.923±.003	0.857±.012	0.205±.001	0.825±.005
■ DESeq2 + Stouffer	Low–bulk–high	0.953±.003	0.917±.005	0.923±.003	**0.869±.012**	0.239±.001	0.812±.005
■ limma-voom + Stouffer	Low–bulk–high	0.954±.003	0.917±.004	0.924±.003	0.862±.011	0.214±.003	0.822±.004
■ BARCS-ST	Low–high	0.941±.004	0.887±.008	0.923±.004	0.615±.013	0.041±.001	0.749±.010
■ BARCS-MOD	Low–high	0.955±.003	0.920±.004	0.923±.004	0.776±.012	0.063±.003	0.849±.006

At genome scale the tradeoff sharpens into a three-way split rather than a single winner. MAGeCK-MLE is the best ranker: it leads AUROC, average precision, and effect recovery. The three general count models recover the most active genes at the nominal threshold, 0.857–0.869 directional recall. BARCS-MOD leads F1 at the nominal threshold, and it is the only method that combines competitive ranking with a realized FDP below the nominal target. The threshold scan below shows that this F1 lead is a property of the nominal operating point rather than of the achievable maximum.

The reason the ranking leader is not the F1 leader is that MAGeCK-MLE converts almost none of its ranking into calls at gene FDR 0.10. Its realized FDP is 0.006, sixteen times below the 0.10 it was allowed, and its directional recall is 0.563 against 0.774 for BARCS-MOD. Paired over the three seeds its average-precision advantage over BARCS-MOD is 0.0024 (95% interval -0.0029 to 0.0077, not resolvable at MOI 0.20), while its F1 deficit is 0.129 (-0.180 to -0.078). It is an accurate but extremely conservative caller here. The count models make the opposite error: their extra recall is bought at realized FDP 0.205–0.239, two to more than two times the nominal target, and it still leaves them behind BARCS-MOD on F1 by 0.022 to 0.035, every interval excluding zero.

BARCS-MOD therefore occupies the operating point the threshold is meant to deliver. It spends most of its FDR budget without exceeding it, 0.066 against the nominal 0.10, and turns that into the highest F1 of any method compared. Its negative-control gene $p < 0.05$ frequency is 0.053 against 0.125–0.135 for the count models. MAGeCK-MLE cannot be scored on that diagnostic: it omits negative-control genes from its gene summary once their sgRNAs are declared, which is also why the common gene set here is 9,475–9,517 rather than 10,000.

The gain over the historical statistic is unambiguous and is a gain in ranking rather than a threshold effect: average precision rises from 0.902 to 0.919, a paired difference of 0.017 (0.011 to 0.022), and F1 from 0.811 to 0.847, with realized FDP statistically unchanged. This is the expected consequence of estimating the moderation prior from 50,000 guides: a guide whose own seven residual degrees of freedom understate its variance is corrected individually, so the negative-control calibration no longer has to answer those guides with one penalty applied to the whole screen.

Bulk does not supply a new directional contrast. Across seeds, the low–bulk–high and tail-only gene effects have mean Spearman correlation above 0.998. For the historical statistic its practical contribution is to the fitted baseline, dispersion, and residual degrees of freedom: mean directional recall rises from 0.615 to 0.716 and F1 from 0.749 to 0.811, at 50% more reads. Moderation largely removes that dependence. BARCS-MOD reaches F1 0.849 on the two tails alone against 0.847 with bulk added, so the information the bulk sample was supplying to the variance model is recovered more cheaply by borrowing it across the library. Because the bulk pool overlaps the tails, its apparent inferential gain may in any case partly reflect an independence approximation rather than new biological information; that reading is strengthened by its being replaceable.

Robustness of FDR control across cutoffs. The realized-FDP column of Table 6 is a single point, and 0.10 is in any case looser than a genome-wide screen would use. We therefore repeated the calibration check on a log-spaced grid of thirteen cutoffs from 10^{-6} to 0.20 in every run. The quantity of interest is the ratio of realized FDP to the cutoff requested: one is exact calibration, below one is conservative, and above one means the method returns more false discoveries than it advertised.

The two-fold overshoot the count models show at 0.10 is the mild end of their behaviour. Their realized FDP stops descending below a requested 10^{-4} and flattens near 5×10^{-4}, so asking for 10^{-6} returns roughly 430 to 560 times the advertised rate. They have an error floor, and no choice of cutoff reaches beneath it. This matters more than the behaviour at 0.10 because it is the regime in which a genome-wide screen is read: a short, confident hit list is exactly what the strict cutoffs are for. Both BARCS statistics instead track the requested rate down to 10^{-4} and return no false discoveries at all below it, holding 37 of 39 cells; MAGeCK-MLE holds all 39 but never spends more than a tenth of its budget.

The BARCS exceptions are worth naming rather than rounding away. Both statistics miss 2 of 39 cells, all from one seed (20250725) at the 3×10^{-4} and 10^{-3} cutoffs, and each miss is a single false discovery against an allowance below one gene. These are Monte Carlo granularity, not a systematic

Table 7: Realized FDP divided by the requested gene-FDR cutoff, MOI 0.20, means over three seeds on the common gene set; a decade-spaced subset of the scanned grid is shown. One is exact calibration and values above one are violations. The final column counts the cutoff-by-seed cells, of $13 \times 3 = 39$, in which realized FDP did not exceed the cutoff.

Method	10^{-6}	10^{-5}	10^{-4}	10^{-3}	0.01	0.05	0.10	Cells held
■ BARCS-ST	0.0	0.0	0.0	0.6	0.6	0.6	0.7	37/39
■ BARCS-MOD	0.0	0.0	0.0	0.4	0.7	0.6	0.7	37/39
■ MAGeCK-MLE	0.0	0.0	0.0	0.0	0.1	0.1	0.1	**39/39**
■ edgeR-QL + Stouffer	556.5	43.7	17.2	10.0	4.1	2.5	2.0	9/39
■ DESeq2 + Stouffer	431.8	105.7	51.7	17.2	5.9	3.2	2.4	5/39
■ limma-voom + Stouffer	531.6	42.1	20.2	10.9	4.8	2.7	2.1	7/39

failure, and they are the reason the calibration claim in this paper is stated as a screen average rather than a finite-sample guarantee. At MOI 0.30 the same pattern holds, with BARCS-MOD holding 35 of 39.

F1 across cutoffs, and why the F1 comparison is fragile. The same scan reports F1 at every cutoff, and doing so exposes a weakness in the single-threshold F1 column that we state rather than work around. *The ordering of methods by F1 is not robust to the cutoff, and it reverses over the strict range.*

Table 8: F1 across requested gene-FDR cutoffs, MOI 0.20, means over three seeds. The leader changes with the cutoff: DESeq2 from 10^{-6} to 0.01, edgeR-QL at 0.05, BARCS-MOD at 0.10.

Method	10^{-6}	10^{-5}	10^{-4}	10^{-3}	0.01	0.05	0.10
■ BARCS-ST	0.046	0.126	0.262	0.456	0.653	0.774	0.811
■ BARCS-MOD	0.248	0.354	0.479	0.607	0.748	0.827	**0.847**
■ MAGeCK-MLE	0.306	0.306	0.306	0.306	0.403	0.622	0.718
■ edgeR-QL + Stouffer	0.474	0.570	0.665	0.757	0.834	**0.846**	0.825
■ DESeq2 + Stouffer	**0.576**	**0.647**	**0.722**	**0.795**	**0.847**	0.841	0.812
■ limma-voom + Stouffer	0.491	0.583	0.675	0.763	0.834	0.844	0.822

BARCS-MOD holds the top F1 only at 0.10. Everywhere stricter, DESeq2 leads, by 0.33 of F1 at 10^{-6}. A reader running a stricter screen would draw the opposite conclusion from Table 6.

The reason is that a fixed requested cutoff does not put these methods at the same place. Table 7 shows DESeq2 running at 430 times its requested rate at 10^{-6}, so it is really operating near a realized 4×10^{-4} and calling 787 genes where BARCS-MOD calls 276. Comparing F1 there compares a liberal operating point with a conservative one, and the comparison says more about where each method sits than about how well it ranks.

Matching on realized FDP instead removes the confound, and F1 then does not separate the methods:

From a matched realized FDP of 0.005 upward the methods converge, to within 0.058, 0.025, 0.019, 0.005, and 0.003 of one another, and which one is nominally highest changes from row to row. We therefore do not claim that guide-dispersion moderation recovers more genes than the alternatives at a common error rate: above 0.005 it does not, and neither does anything else here. Below that the spread widens again, to 0.202 at a matched 0.001, but that region is thinly covered: several methods reach a realized 0.001 only by extrapolating between widely separated grid points,

Table 9: F1 at matched realized FDP, MOI 0.20, interpolated within each run before averaging over the three seeds. A dash marks a target outside the realized-FDP range a method attains on the scanned grid. Spread is the range across the methods other than BARCS-ST.

Method	0.001	0.002	0.005	0.01	0.02	0.05	0.10
■ BARCS-ST	0.491	0.529	0.631	0.678	0.738	0.795	0.819
■ BARCS-MOD	0.640	0.677	0.724	0.764	0.808	0.838	0.847
■ MAGeCK-MLE	0.469	0.509	0.666	0.739	0.789	—	—
■ edgeR-QL + Stouffer	0.664	0.626	0.709	0.757	0.796	0.836	0.845
■ DESeq2 + Stouffer	0.671	0.653	0.717	0.761	0.801	0.840	0.849
■ limma-voom + Stouffer	0.656	0.637	0.707	0.759	0.801	0.835	0.848
Spread	0.202	0.168	0.058	0.025	0.019	0.005	0.003

and MAGeCK-MLE's quantized FDR makes its entry there unreliable. We read the convergence above 0.005 as the robust statement and treat the two strictest rows as indicative.

Two claims do survive the scan. First, the improvement of BARCS-MOD over BARCS-ST is robust to the cutoff, being larger at every matched realized FDP, so moderation is a genuine gain within the BARCS family rather than a threshold artifact. Second, and this is the substantive result, BARCS-MOD is the method whose requested cutoff lands where the F1 curve is already flat and high while the realized error rate is still below what was asked for. F1 does not discriminate between these methods; what discriminates them is whether the number you set is the number you get.

MAGeCK-MLE cannot be compared at strict cutoffs, and more permutations do not fix it. MAGeCK-MLE's call count is constant across several orders of magnitude of requested FDR, which is a property of its inference rather than of the data. Its gene p-value is a permutation tail probability estimated from genes×rounds draws, so it is quantized in steps of about 1/(genes×rounds), and any gene whose statistic exceeds every permuted value is reported with p exactly zero. Such genes are called at any cutoff however strict, which is what produces the plateau.

Refitting one realization at ten rounds instead of one confirms the mechanism and shows that it is not a settings artifact that can simply be corrected. The smallest positive p-value falls from 2×10^{-4} to 2×10^{-5}, matching the predicted $1/(9475 \times \text{rounds})$ to within the two-sided factor, and the number of genes at $p = 0$ falls from 186 to 66. But the plateau does not disappear: with ten rounds the call count is still flat, at 66 genes from 10^{-6} through 10^{-3}, and the F1 over that range *falls* from 0.174 to 0.065 because fewer genes reach $p = 0$. More permutation moves the plateau down rather than removing it, so MAGeCK-MLE's strict-cutoff position is set by a computational budget rather than by the screen.

Resolving a gene FDR of 10^{-6} at this library size would need on the order of $10^6/9475 \approx 110$ rounds, roughly fourteen hours per realization at the throughput measured here. We therefore read MAGeCK-MLE's curve below a requested 10^{-2} as not resolvable rather than as a performance result, and exclude it from the strict-regime comparison; the same caution applies to any permutation-based gene FDR used at genome scale. This is measured in `examples/crispulator_facs_mageck_permutation_resolution.R`.

At MOI 0.30 every one of these conclusions is reproduced (Supplementary Table 10): BARCS-MOD again leads F1 at FDP 0.070, the count models again run at 0.200–0.236, and MAGeCK-MLE again ranks marginally best while calling least, at FDP 0.003 and F1 0.708. The threshold scan repeats too: peak F1 spans 0.825–0.855 across methods, and BARCS-MOD again attains its own peak at the nominal 0.10 while MAGeCK-MLE gives up 0.146 there. The one difference is that

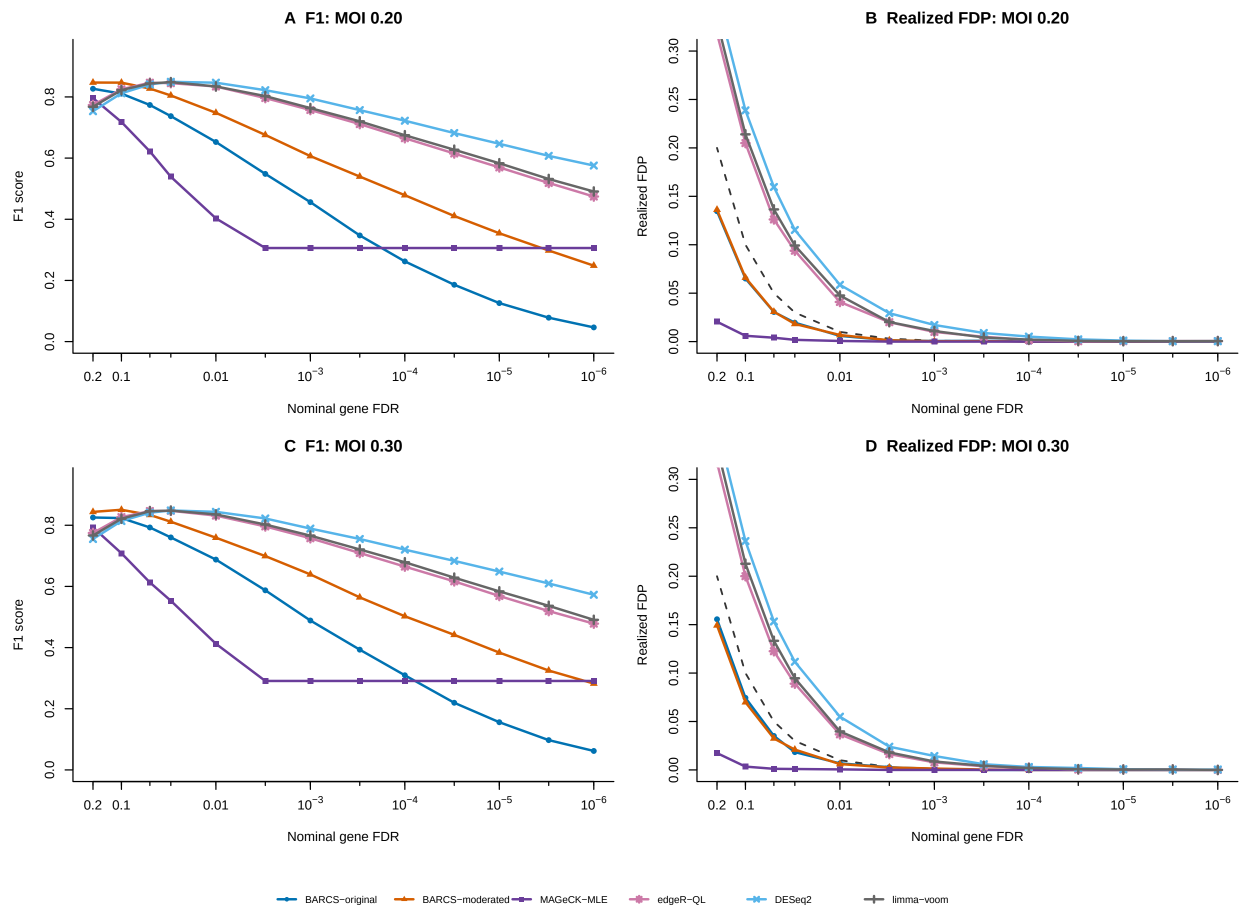

Figure 4: Calibration and F1 across gene-FDR cutoffs in the genome-scale simulation, means over three seeds on the common gene set. (A) F1 and (B) realized FDP at MOI 0.20; (C) and (D) the same at MOI 0.30. The dashed line in the FDP panels is realized equal to nominal, so a method tracking it reports the error rate it delivers. Thresholds above 0.10 are a diagnostic: they show what each method could attain if its threshold were chosen with knowledge of the truth, which a real screen does not have.

MAGeCK-MLE's average-precision margin over BARCS-MOD is resolvable at this MOI, 0.0033 (0.0017 to 0.0049).

Supporting 400-gene sensitivity to MOI, guide quality, library size, and replication. The remaining simulation analyses in this subsection are the earlier 400-gene grid, retained because they map the parameter boundaries that the genome-scale benchmark deliberately does not vary, including the one-replicate boundary. They predate the moderated statistic and use the historical BARCS-ST default and the earlier MOI $\in \{0.10, 0.25, 0.40\}$ levels, so they should be read as boundary mapping rather than as the headline comparison. We varied MOI, the high-quality-guide fraction, the number of genes, and the number of independent screen replicates one at a time, using five paired seeds at every setting (ten unique settings and 50 simulations in total). Nine settings had at least three replicates; the one-replicate setting was a prespecified diagnostic boundary. Among the supported settings, in the primary low–bulk–high comparison, BARCS-minus-MAGeCK-MLE average precision

Table 10: Supplementary: the same genome-scale CRISPulator FACS benchmark at MOI 0.30 rather than 0.20, over the same three seeds (mean ± standard deviation). A higher multiplicity of infection stresses the single-integration approximation shared by every guide-level model compared here. Scoring and highlighting follow Table 6.

Method	Samples	AUROC	Avg. precision	Effect ρ_s	Directional recall	FDP	F1
BARCS-ST	Low–bulk–high	0.947±.001	0.903±.003	0.923±.003	0.742±.007	0.074±.009	0.823±.001
BARCS-MOD	Low–bulk–high	0.955±.002	0.920±.003	0.923±.003	0.784±.008	0.070±.004	**0.851±.003**
MAGeCK-MLE	Low–bulk–high	**0.959±.002**	**0.923±.003**	**0.932±.001**	0.550±.009	0.003±.002	0.708±.008
edgeR-QL + Stouffer	Low–bulk–high	0.954±.002	0.917±.002	0.922±.003	0.855±.002	0.200±.003	0.826±.001
DESeq2 + Stouffer	Low–bulk–high	0.954±.002	0.918±.003	0.921±.003	**0.871±.003**	0.236±.009	0.814±.004
limma-voom + Stouffer	Low–bulk–high	0.954±.002	0.918±.003	0.923±.003	0.858±.007	0.213±.003	0.821±.004
BARCS-ST	Low–high	0.941±.001	0.888±.002	0.923±.003	0.616±.012	0.033±.004	0.752±.009
BARCS-MOD	Low–high	0.956±.002	0.921±.003	0.923±.003	0.778±.003	0.060±.003	0.851±.001

ranged only from −0.015 to 0.022 and changed sign across settings. Thus this experiment does not support a general ranking advantage for BARCS. Its more consistent strength was directional recovery at the prespecified FDR threshold: the directional-recall difference was positive in seven of nine settings and ranged from −0.018 to 0.136, while the F1 difference ranged from −0.022 to 0.080.

The largest directional-recall differences occurred at MOI 0.10 (+0.107) and a high-quality-guide fraction of 0.75 (+0.119). At MOI 0.40, however, MAGeCK-MLE led directional recall by 0.018 and F1 by 0.022; with only 60% high-quality guides, the methods were nearly tied on directional recall and MAGeCK-MLE led F1 by 0.005. Across 100, 400, and 1,000 genes, BARCS directional-recall differences were +0.081, +0.047, and +0.064, respectively, whereas average-precision differences again remained small. The 100-gene setting also produced realized FDP above 0.10 for both methods (0.181 for BARCS and 0.147 for MAGeCK-MLE), warning that five small-library replicates do not establish finite-sample FDR control. These results sharpen the claim: BARCS can improve directionally correct thresholded recovery under some noisy or low-MOI conditions, but neither method dominates every metric or simulation regime.

Replication changed that comparison. With three replicates, BARCS exceeded MAGeCK-MLE in directional recall by 0.136 and F1 by 0.080; with four replicates the differences were 0.047 and 0.011. At six replicates the signs reversed slightly, to −0.012 for directional recall and −0.014 for F1. The practical interpretation is not that fewer replicates are desirable: average precision, recall, and F1 increased for both methods as replication increased. Rather, BARCS's relative sensitivity is largest in the low-replication regime, whereas MAGeCK-MLE catches up when information is abundant.

One replicate crossed from low information into non-confirmatory inference. Ordinary BARCS retained higher average precision than MAGeCK-MLE, 0.633 versus 0.548, but made no discoveries at gene FDR 0.10 across the five seeds; MAGeCK-MLE's mean directional recall was only 0.014. The new BARCS-GC guide-consistency statistic increased average precision to 0.670, directional recall to 0.145, and F1 to 0.245. It made 58 calls across the five simulations (3–19 per seed), all of which were active genes under the simulated truth, for mean realized FDP 0. The fitted empirical-null scales were 2.19–2.60, showing why an unscaled Wald reference would have been anti-conservative.

This improvement is useful but deliberately narrow. BARCS-GC asks whether independently designed guides give concordant signed effects and assumes that fewer than half of genes are active; it does not create biological replication. Its empirical-null calls are therefore hypothesis-generating, not confirmatory. edgeR-QL and limma-voom reported mean directional recall of 0.708 and 0.684 in the same data, but their mean realized FDP was 0.526 and 0.507. DESeq2 was less inflated (FDP 0.141), yet still exceeded the nominal 0.10 target. We recommend an independent screen for confirmation

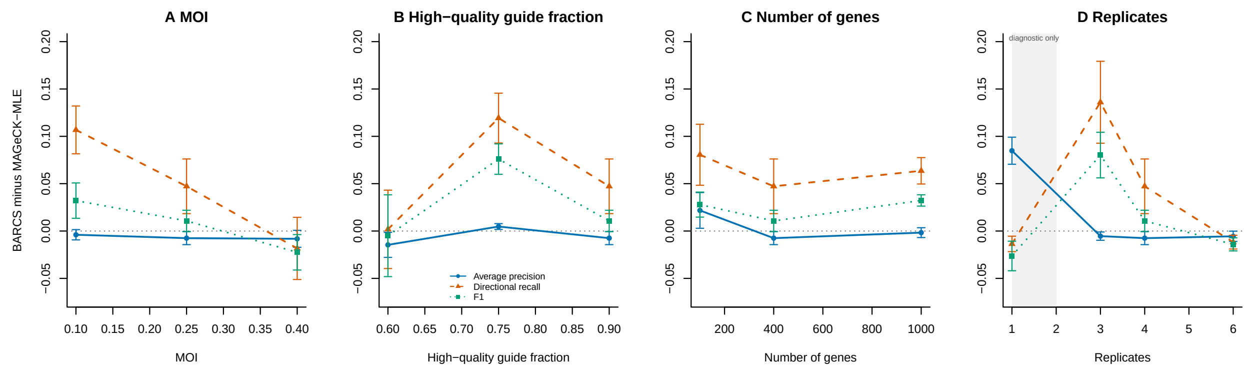

Figure 5: Paired parameter sensitivity across five seeds per setting. Points show mean BARCS-minus-MAGeCK-MLE differences in the low–bulk–high design, with standard-error bars. Positive values favor BARCS. Each panel varies (A) MOI, (B) high-quality-guide fraction, (C) number of genes, or (D) independent screen replicates while holding the other parameters at their baseline values. Ranking differences remain close to zero, whereas directional-recall differences are usually positive but reverse at MOI 0.40 and six replicates. The shaded $R = 1$ region is a diagnostic boundary and not a recommended confirmatory design.

even when BARCS-GC is used to prioritize genes from an otherwise irreplaceable $R = 1$ experiment.

Multiple count models expose the calibration–power tradeoff. Across the nine supported scenarios with $R \geq 3$, mean realized FDP was 0.094 for BARCS and 0.064 for MAGeCK-MLE, compared with 0.250 for edgeR-QL plus Stouffer, 0.243 for DESeq2 plus Stouffer, and 0.230 for limma-voom plus Stouffer. Figure 6 makes the consequence visible. The general count models attain excellent average precision and raw directional recall, but their apparent recall advantage is coupled to FDP near 0.20–0.25. Restricted to these supported settings, their F1 advantage over BARCS is not resolvable: the paired differences are +0.008 for edgeR-QL (95% interval −0.015 to +0.030), +0.013 for DESeq2 (−0.010 to +0.035), and +0.014 for limma-voom (−0.008 to +0.037), whereas the paired average-precision advantage of 0.016–0.017 remains clear. At six replicates, BARCS and MAGeCK-MLE reach F1 of 0.893 and 0.907 while keeping mean FDP at 0.079 and 0.064; edgeR-QL, DESeq2, and limma-voom reach F1 of 0.840–0.862 with FDP of 0.196–0.235. This is evidence for the value of a CRISPR-aware gene-testing and calibration pipeline, not evidence that the general count models rank genes poorly.

At the four-replicate baseline, mean core fitting times were 0.35 seconds for BARCS, 4.34 seconds for MAGeCK-MLE, 0.24 seconds for edgeR-QL, 0.92 seconds for DESeq2, and 0.06 seconds for limma-voom. These measurements exclude simulation, file input/output, and guide-to-gene aggregation; they show that the broader comparison is computationally practical but should not be read as an end-to-end software benchmark.

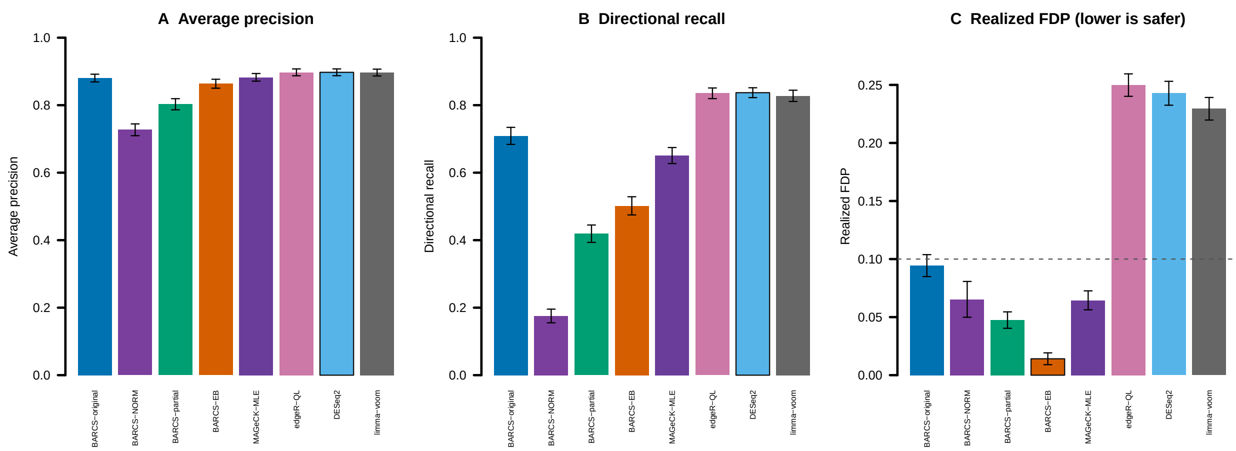

Figure 6: Head-to-head comparison across the 45 multi-replicate low–bulk–high CRISPulator runs. The one-replicate setting is a prespecified diagnostic boundary and is excluded here; it is reported separately in the text. Bars are means and error bars are standard errors. (A) Average precision, (B) directionally correct recall at gene FDR 0.10, and (C) realized FDP, where lower is safer and the dotted line is the nominal 0.10 target. edgeR-QL, DESeq2, and limma-voom use the common directional Stouffer gene summary; MAGeCK-MLE uses its native gene model.

2.5 Backward compatibility: GSE70038

GSE70038 contains 64,747 guides measured in 16 libraries from two glioblastoma stem-cell lines and two neural stem-cell lines, with duplicate initial and terminal samples [9]. This binary terminal-versus-initial setting is a special case of the regression model, not a continuous-phenotype benchmark. It is nevertheless useful because it provides published biology and permits an exact design-matched comparison with MAGeCK-MLE.

We followed the design-matrix rules in Table 5 and Box 3 of the MAGeCKFlute protocol [10]. The first column is one for every library; each initial library has zeros elsewhere; and each terminal library has one in exactly one of four cell-line columns. The same 16×5 matrix was supplied to beta-binomial regression and the official MAGeCK-MLE 0.5.9.5 executable. MAGeCK used median normalization and its Wald inference. For the beta-binomial result, guide effects were summarized by their within-gene median, and directional guide p-values were combined by Stouffer's method. This aggregation is an explicit comparison device, not a proposed replacement for MAGeCK's gene model.

The "effect rank correlation" reported below is calculated in this analysis, not copied from GEO or the MAGeCKFlute paper. Separately for each terminal coefficient, we pair all genes available from both methods. The beta-binomial effect is the within-gene median guide coefficient, and the MAGeCK effect is its official MLE beta. We assign average ranks to these two effect vectors and take their Pearson correlation, which is exactly Spearman's rank correlation. The 18,077 paired effects and their ranks for every coefficient are written to `results/gse70038/effect_concordance_pairs.csv.gz`.

Table 11: Effect concordance between ■ BARCS and ■ official MAGeCK-MLE on GSE70038. Jaccard is the overlap divided by the union of the 200 most depleted genes from each method. Higher agreement is better; column maxima are shown in bold.

Terminal coefficient	Spearman effect correlation	Top-200 Jaccard
GSC0131	**0.928**	0.550
GSC0827	0.888	0.533
NSCCB660	0.905	0.544
NSCU5	0.915	**0.562**

The comparison is not circular: beta-binomial regression conditions on each library total, whereas MAGeCK uses a normalized negative-binomial count model with guide-efficiency estimation [11]. Despite those differences, all four effect correlations exceed 0.88. These correlations measure agreement of the inferred gene ordering; they are not likelihood, calibration, or predictive-fit statistics and therefore cannot identify a better-fitting method. In GSC0131, the published convergent dependency PKMYT1 has effect -0.987 and FDR 1.87×10^{-8} under beta-binomial regression, versus beta -0.786 and Wald FDR 4.53×10^{-8} under MAGeCK-MLE. Both also recover HEATR1, TFAP2C, and WEE1 depletion. Figure 7 shows the complete GSC0131 comparison; the reproducible PDF contains one page per terminal coefficient.

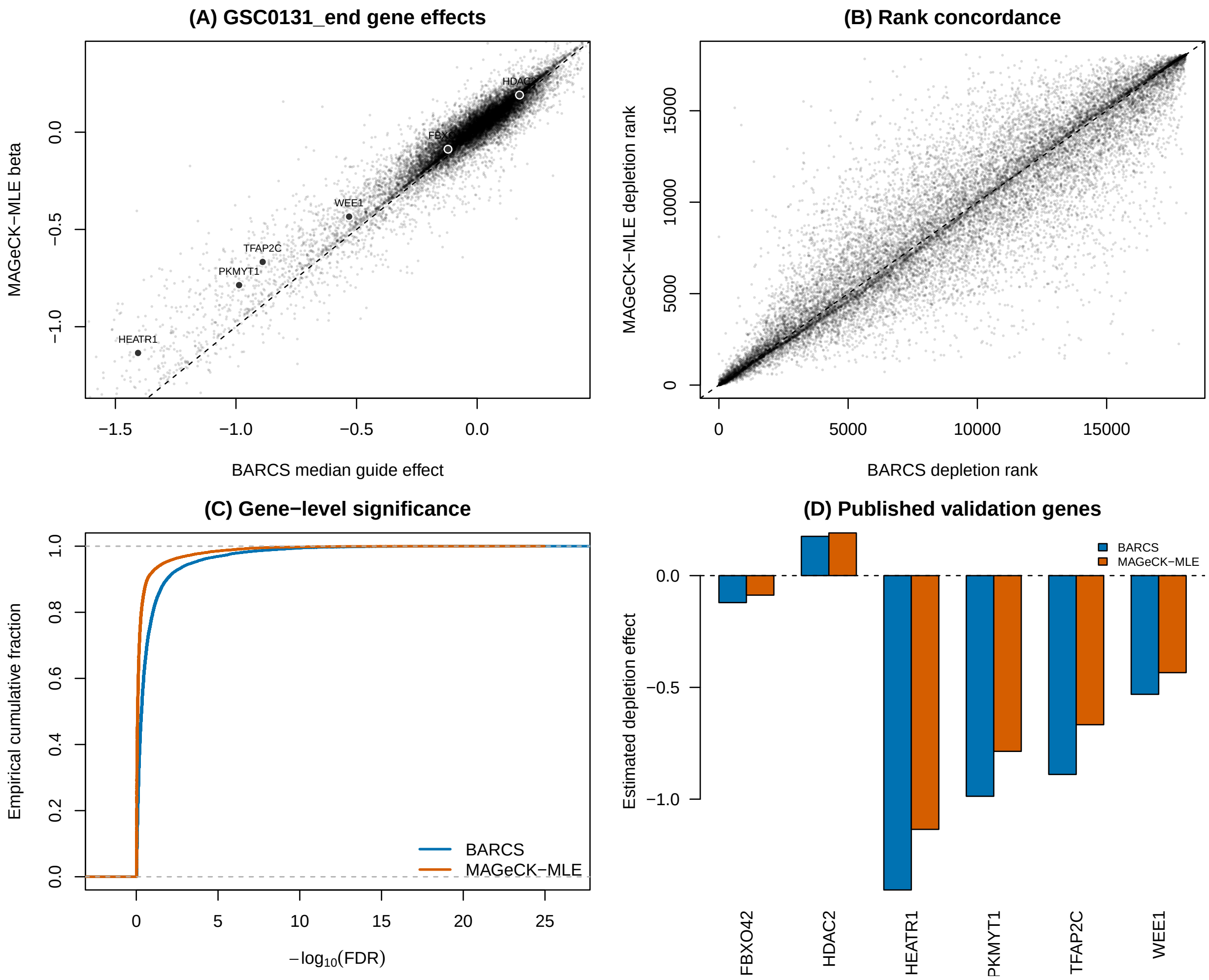

Figure 7: GSE70038 GSC0131 head-to-head. (A) Gene effects, with dark points marking the prespecified genes PKMYT1, HEATR1, TFAP2C, FBXO42, HDAC2, and WEE1. (B) Depletion-rank concordance. (C) Gene-level FDR distributions. (D) Effects for the published validation genes.

2.6 Endpoint benchmark and copy-number audit

To ask which method ranks biological dependencies better, we used the Brunello A375 negative-selection screen of Sanson et al. [12], bundled with CB^2. This is an ordinary endpoint screen rather than a continuous-phenotype experiment, but it supplies an external target unavailable in GSE70038: reference essential genes should rank ahead of reference nonessential genes. The beta-binomial score combines guide evidence by directional Stouffer aggregation; the MAGeCK score uses the official MLE beta and Wald p-value. Both are oriented so larger scores mean stronger depletion.

2.6.1 Copy-number-aware analysis

Copy number is a gene-level property of A375, not a sample-level covariate: for a given guide it is constant across the plasmid and three terminal libraries. Adding it directly to the four-row sample design would therefore be non-identifiable with the endpoint effect. Instead, we downloaded the A375 gene-level profile (ACH-000219) from DepMap Public 19Q3 [13, 14] and used MAGeCK 0.5.9.5's official `-cnv-norm` piecewise correction. For an apples-to-apples comparison, the same MAGeCK piecewise normalizer was applied to the beta-binomial within-gene median effects. It adjusts effect sizes after model fitting; it does not refit either count likelihood. The common CNV-complete evaluation set contains 1,506 essential and 869 nonessential genes.

Table 12: CNV-aware Sanson A375 benchmark. The final column is Spearman correlation between effect and copy number among reference nonessential genes; values nearer zero indicate less residual CNV-associated depletion. F1 uses nominal FDR 0.05 and a negative effect. Higher discrimination metrics and CNV correlation closest to zero are highlighted.

Method	AUROC	Avg. precision	F1	CNV correlation
■ BARCS	**0.9598**	0.9786	**0.8531**	−0.185
■ BARCS + CNV	0.9533	0.9762	**0.8531**	−0.042
■ Official MAGeCK-MLE	**0.9598**	**0.9795**	0.8277	−0.207
■ Official MAGeCK-MLE + CNV	0.9501	0.9762	0.8277	**−0.006**

CNV correction substantially removes the intended nuisance association: −0.185 to −0.042 for beta-binomial effects and −0.207 to −0.006 for MAGeCK. It does not improve essential-gene discrimination in this screen. AUROC decreases by 0.0066 and 0.0097, respectively, and the 0.05-FDR calls are unchanged. This is not a contradiction: bias removal and gold-standard classification are different targets, and amplified genes can include both dependencies and nondependencies.

After CNV correction, beta-binomial versus MAGeCK global ranking remains indistinguishable: the AUROC difference is 0.00316 (paired stratified-bootstrap 95% CI −0.00265 to 0.00924), and the average precision difference is effectively zero (95% CI −0.00349 to 0.00295). At nominal FDR 0.05, beta-binomial retains 4.18 percentage points more recall (95% CI 2.66 to 5.71) and an F1 advantage of 0.0253 (95% CI 0.0148 to 0.0362).

2.6.2 Why beta-binomial does not dominate MAGeCK

The A375 screen is favorable to MAGeCK: it is a binary endpoint experiment, MAGeCK's native use case, with one plasmid baseline and three terminal libraries. A guide-wise beta-binomial fit then has only two residual degrees of freedom, making its t tail deliberately conservative. MAGeCK instead borrows information across the screen, estimates guide efficiency, and uses asymptotic Wald inference. These features explain its advantage at very small FDR thresholds; the intended advantage of using a genuinely continuous predictor is not exercised by this dataset.

MAGeCK is nevertheless not an oracle. Its negative-binomial mean–variance model, median normalization, guide-efficiency model, and normal-reference Wald tests are assumptions rather than guarantees [11, 15]. They can be sensitive to global composition changes, atypical guide behavior, or limited biological replication. Moreover, MAGeCK 0.5.9.5 performs CNV normalization after its permutation p-values and FDR values are calculated. Consequently, CNV correction can remove effect–CNV association while leaving the significance calls unchanged, exactly as observed here. Conversely, the present beta-binomial prototype estimates dispersion guide by guide and uses a transparent but ad-hoc Stouffer gene aggregation. A fair path to more consistent gains is empirical-Bayes dispersion moderation plus a native gene-level hierarchy, not changing the benchmark or weakening MAGeCK.

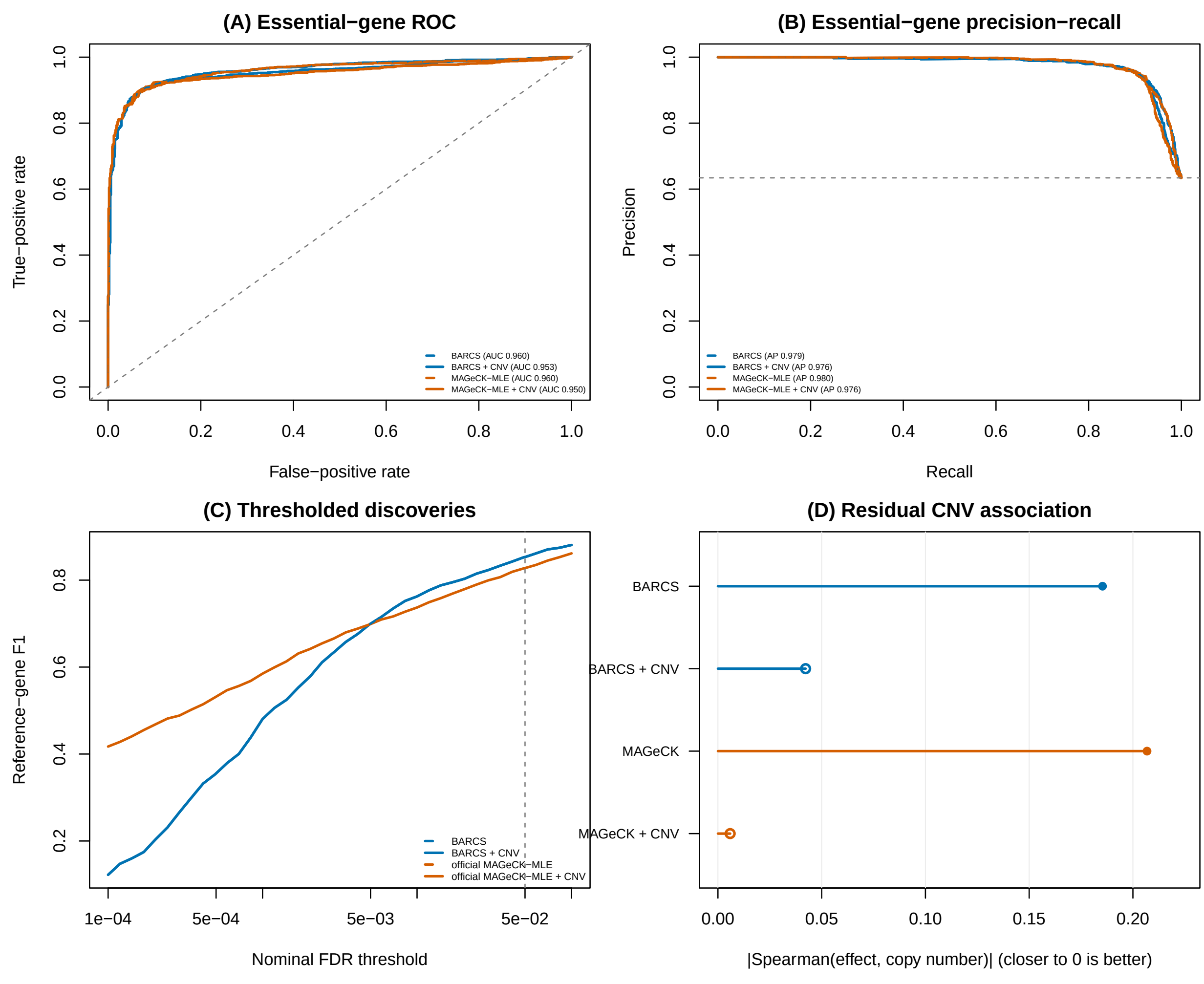

Figure 8: CNV-aware essential-gene discrimination in the Sanson A375 screen. (A) ROC, (B) precision–recall, (C) F1 across nominal FDR thresholds, and (D) absolute residual effect–CNV correlation among nonessential genes, where values closer to zero are better. Curves compare uncorrected (dashed) and CNV-corrected (solid) BARCS and official MAGeCK-MLE results; open points mark CNV-corrected values.

2.7 External false-discovery benchmark: sensitivity to the analysis contract

A recent Chronos hit-calling preprint compared CB^2, MAGeCK, JACKS, DrugZ, and Chronos on single-condition dependency calling and differential viability [16]. The deposited notebooks and intermediate files make this comparison unusually auditable [17]. They expose two distinct observations that should not be conflated. Original CB^2 has a genuine small-sample power limitation in two-versus-two condition comparisons, but the strongest reported evidence of anti-conservative CB^2 inference is sensitive to an incompatible preprocessing step.

For guide g and sample j, CB^2 constructs the library total internally as

$$N_j = \sum_{g \in \mathcal{G}} K_{gj}.$$

The null-only Avana benchmark first restricted the count matrix to selected control genes and then ran CB^2. Consequently, its beta-binomial denominator became

$$N_j^{\text{subset}} = \sum_{g \in \mathcal{G}_0} K_{gj},$$

rather than the full sequencing depth. Filtering guides is legitimate for evaluation, but it must occur after fitting or be accompanied by immutable full-library totals. This distinction matters because within-sample rescaling leaves guide proportions unchanged while changing the beta-binomial sampling variance through N_j.

We audited this point using the deposited `CB2_no_positives.csv` and `CB2_allgenes.csv` files. The published subset-matrix analysis reports 152 test-negative discoveries at FDR 0.1. Restricting the full-library CB^2 p-values to exactly the same test genes and reapplying Benjamini–Hochberg gives zero discoveries in all five cell lines (Table 13). The nominal $p < 0.05$ frequency still reaches 14.5% and 11.7% in two lines, so this correction does not establish perfect calibration. It does show that the reported 152 discoveries are not evidence about CB^2 fitted with its required full-library denominator. A second concern is that the common preprocessing applies Chronos negative-control normalization and rounding before all fits. That transformation is meaningful for Chronos, but it changes the effective sequencing depths of a method whose likelihood treats read totals as sampling information. Equal preprocessing is therefore not automatically equivalent to equal model input.

Table 13: Audit of the version-1 Chronos false-discovery benchmark. The Avana row uses the same test genes under two library-total contracts. The PSN1 and DeWeirdt rows are direct BARCS reruns on the deposited normalized count tables; "known" denotes the 28 literature-curated condition–gene combinations used by the preprint, not a complete truth set.

Audit target	■ Reported legacy result	■ BARCS reanalysis
Avana null-only, FDR < 0.1	152 false discoveries after pre-fit guide subsetting	0 after preserving full-library totals
PSN1 two-versus-two null split	0 gene discoveries; conservative gene p-values	4.81% guide $p < 0.05$, 4.65% after control scaling, and 0 gene discoveries
DeWeirdt differential screens	0 discoveries and 0/28 known interactions	38 discoveries and 1/28 known interactions

The differential-screen limitation is real. Original CB^2 estimates each condition separately and uses the Welch–Satterthwaite degrees of freedom

$$\nu = \frac{(\widehat{V}_A + \widehat{V}_B)^2}{\widehat{V}_A^2/(n_A - 1) + \widehat{V}_B^2/(n_B - 1)}.$$

With $n_A = n_B = 2$, $1 \le \nu \le 2$. Two independent variance estimates and a heavy-tailed reference distribution leave little power. In contrast, BARCS estimates one guide-level dispersion across the full design. On the PSN1 pseudo-condition split, its guide-level null is close to nominal and its negative-control scale is 1.017. This is evidence that pooled regression repairs part of the finite-sample problem.

It does not solve the full Chronos task. In an exploratory rerun of the ten DeWeirdt comparisons used for the curated-positive analysis [18], BARCS used the four terminal libraries, full count-table totals, a treatment coefficient, guide-level control scaling, Fisher gene aggregation, and BH correction. It recovered MCL1 under A-1331852 in Meljuso, but only one of 28 curated interactions overall. Thirty-seven of its 38 discoveries came from OVCAR8 under olaparib, without recovering BRCA1 or BRCA2. This pattern is consistent with a global difference in screen sensitivity rather than a failure that one additional fixed covariate can remove: with two replicates per arm, a treatment-associated quality shift is nearly collinear with treatment itself. Negative controls that are inactive in both arms also need not represent artifacts affecting strong common-essential knockouts.

The version-1 notebook raises two further reproducibility issues. First, its FDR-calibration helper accepts a requested evaluation gene set but does not use that argument; the calibration curve therefore includes training controls used by Chronos as well as held-out controls. Second, calls intended to remove MAGeCK's `common` coefficient use a non-mutating `DataFrame.drop()` without assigning the returned object. In the PSN1 calibration, this leaves 34,124 MAGeCK entries versus 17,787 entries for CB^2, JACKS, DrugZ, and Chronos. These implementation details do not invalidate the biological motivation for empirical-null hit calling, but they make the strongest method-ranking and calibration claims provisional. Similarly, treating thousands of correlated gene–screen indicators as independent observations in a t test yields uncertainty that is narrower than a screen- or cell-line-clustered bootstrap would provide.

The appropriate conclusion is therefore narrower than either "CB^2 is miscalibrated" or "BARCS solves false discovery." The regression extension repairs the mean-model and dispersion-pooling limitations, accepts CNV and other identifiable sample covariates, and gives a well-behaved null in the PSN1 check. It does not by itself supply Chronos's growth dynamics, replicate-quality terms, or empirical differential null. A production differential-screen method still needs immutable library totals, raw-count or likelihood-compatible normalization, dispersion moderation, correlation-aware gene aggregation, held-out control calibration, and blocked replicate-label permutations.

3 Methods

3.1 From CB^2 to BARCS

BARCS is fundamentally a guide-level coefficient model. Its required inputs are a guide-by-library count matrix, the corresponding full-library totals, and a sample design matrix; its primary output is one coefficient or contrast test per guide. Gene aggregation is downstream and optional. In particular, neither Fisher nor Stouffer combination is required to fit or interpret the BARCS regression. Some benchmarks below use an explicitly labeled Stouffer summary to compare with deposited gene-level results, whereas BARCS-GC uses the direct shared-effect Wald construction in Equations (28)–(31).

CB^2 adapts the beta-binomial weighted test of Baggerly et al. [19] to CRISPR pooled screens [20]. Its central insight is worth preserving: sequencing depth measures a guide proportion precisely, but it does not turn reads into independent biological replicates. The original method applies this insight to a two-group contrast. BARCS retains the same depth-aware, extra-binomial variance principle and changes the mean model from two group means to a design matrix. “Retains” refers to the stochastic hierarchy and the sample-level information ceiling; it does not mean that the default logit fit reproduces the original finite-sample statistic. The legacy representation and asymptotic relationship are established below.

Table 14: What changes—and what does not—from the original CB^2 method to BARCS.

Feature	Original CB^2	BARCS
Primary question	Difference between two groups	Trend, adjusted association, interaction, or contrast
Sample design	Reference and comparison labels	Arbitrary full-rank R model matrix
Mean model	Two unrelated group proportions	$\operatorname{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$
Effect	Difference or fold change between groups	Conditional log-odds coefficient or linear contrast
Dispersion	Separate beta-binomial weighting within each group	One guide-specific ρ shared across the fitted design
Estimator	Weighted group proportions	Feasible logistic IRLS
Variance principle	Binomial sequencing plus between-library beta-binomial heterogeneity	
Reference df	Group-variance df formula	Residual $m - q$
Adjustment	Pairwise stratification or preprocessing	Batch, donor, dose, time, interactions, splines, and other covariates
Implementation	Existing CB^2 workflow	Additive `bbreg()`, `bb_contrast()`, and `bb_screen()` APIs with RcppArmadillo kernels

The regression bridge was described by Baggerly et al. [21] for sequencing count data with multiple groups and continuous covariates. The method combines logistic regression with extra-binomial variation and retains a coefficient-based t test. Here we state that construction in CRISPR-screen notation and use the beta-binomial, library-size-dependent variance of Williams [22]. An identity-link regression of raw proportions would allow fitted values outside $(0, 1)$, while an unweighted correlation would ignore unequal library sizes. The logit and beta-binomial weights solve those two problems directly.

3.2 Model

For guide g in sequenced sample i, let K_{gi} be its read count, let N_i be the full mapped-guide total for that sample, and let $Y_{gi} = K_{gi}/N_i$ be the observed guide proportion. Sample-level quantities such as dose, time, treatment, donor, and batch are predictors even when they are described biologically as experimental “outcomes.”

The model is fitted separately for every guide, so the guide index is suppressed below. Conditional on a latent sample-specific abundance P_i,

$$K_i \mid P_i \sim \text{Binomial}(N_i, P_i), \tag{2}$$

$$P_i \sim \text{Beta}(\mu_i\kappa, (1 - \mu_i)\kappa). \tag{3}$$

The mean is linked to a row vector of sample covariates $\mathbf{x}_i^\top$ by

$$\operatorname{logit}(\mu_i) = \log\left(\frac{\mu_i}{1 - \mu_i}\right) = \mathbf{x}_i^\top \boldsymbol{\beta}. \tag{4}$$

The design can contain an intercept, a continuous phenotype, batch indicators, interactions, spline terms, and other prespecified adjustment variables.

It is useful to express the beta precision κ as an intraclass correlation

$$\rho = \frac{1}{\kappa + 1}, \qquad 0 \le \rho < 1. \tag{5}$$

Marginally,

$$\begin{aligned} \mathrm{E}(K_i) &= N_i \mu_i, & (6)\\ \mathrm{Var}(K_i) &= N_i \mu_i (1-\mu_i)\{1 + (N_i - 1)\rho\}, & (7)\\ \mathrm{Var}(Y_i) &= \mu_i(1-\mu_i)\left\{\frac{1-\rho}{N_i} + \rho\right\}. & (8) \end{aligned}$$

Equation (8) cleanly separates sequencing uncertainty, which decreases with N_i, from between-sample heterogeneity, which does not vanish when sequencing depth becomes arbitrarily large.

3.2.1 Interpretation of the continuous coefficient

Suppose the model contains a continuous phenotype d_i with coefficient β_d. Then $\exp(\beta_d)$ is the multiplicative change in the odds of observing that guide for a one-unit increase in d_i, conditional on the other covariates. Standardizing d_i before fitting makes $\exp(\beta_d)$ the odds ratio per sample standard deviation and makes guide effects directly comparable. For rare guides, odds and proportions are similar, so β_d is also approximately a log abundance ratio per unit of d_i.

3.3 Estimation by feasible IRLS

Let X be the $m \times q$ full-rank design matrix. Given current values of $\boldsymbol{\beta}$ and ρ, define

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad \mu_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \tag{9}$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1-\mu_i)}, \qquad w_i = \frac{N_i \mu_i (1-\mu_i)}{1 + (N_i - 1)\rho}. \tag{10}$$

The coefficient update is the weighted least-squares step

$$\widehat{\boldsymbol{\beta}}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \qquad W = \mathrm{diag}(w_1, \ldots, w_m). \tag{11}$$

This is weighted least squares on the logistic working response, not on the raw proportions.

For a fixed fitted mean, ρ is estimated from the Pearson equation

$$\sum_{i=1}^{m} \frac{(K_i - N_i \widehat{\mu}_i)^2}{N_i \widehat{\mu}_i (1 - \widehat{\mu}_i)\{1 + (N_i - 1)\widehat{\rho}\}} = m - q. \tag{12}$$

If the left side at $\rho = 0$ is no larger than $m - q$, the estimate is truncated at zero rather than claiming negative overdispersion. Equations (10)–(12) are alternated to convergence.

The weight has an important limiting behavior:

$$\lim_{N_i \to \infty} w_i = \frac{\mu_i(1-\mu_i)}{\rho} \quad \text{when } \rho > 0. \tag{13}$$

Therefore a deeply sequenced sample cannot acquire unlimited leverage. Once sequencing uncertainty is small, biological heterogeneity sets the information ceiling.

3.4 T-based inference

At convergence, the model-based covariance estimator is

$$\widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}}) = (X^\top \widehat{W} X)^{-1}. \tag{14}$$

The Pearson equation scales the residual variation to approximately one. If the dispersion estimate reaches its numerical upper boundary, the implementation additionally multiplies Equation (14) by the remaining Pearson scale.

Equation (14) is a plug-in, model-based covariance: it conditions on the estimated $\widehat{\rho}$ as though it were fixed. It does not propagate uncertainty from estimating a separate dispersion for every guide. Referring the coefficient to t_{m-q} gives a heavier-tailed small-sample reference than a normal approximation, but it is not a formal correction for dispersion-estimation uncertainty. With few independent libraries, calibration must therefore be checked by simulation, phenotype permutation, or prespecified negative controls.

For coefficient β_j, the statistic

$$t_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}})_{jj}}} \tag{15}$$

is compared with a Student t_{m-q} distribution. The degrees of freedom are driven by the number of independent sequenced samples, not by the total read count. This is the crucial reason that a library with millions of reads does not create artificial certainty.

More generally, for a prespecified contrast vector $\mathbf{c}$,

$$t_{\mathbf{c}} = \frac{\mathbf{c}^\top \widehat{\boldsymbol{\beta}} - \delta_0}{\sqrt{\mathbf{c}^\top \widehat{\mathrm{Cov}}(\widehat{\boldsymbol{\beta}})\mathbf{c}}} \tag{16}$$

uses the same $m - q$ degrees of freedom. This covers pairwise group contrasts, average slopes in interaction models, and other one-degree-of-freedom hypotheses. Guide-wise two-sided p-values are adjusted using the Benjamini–Hochberg procedure [23].

3.4.1 From guide statistics to gene statistics

BARCS fits the count model guide by guide. The primary HT-29, FACS, simulation, GSE70038, and A375 benchmark scripts then use an explicit directional Stouffer summary, denoted BARCS-ST below, to obtain one gene-level statistic. For the jth valid guide targeting gene g, the two-sided guide p-value is converted back to a signed standard-normal score,

$$z_{gj} = \mathrm{sign}(\widehat{\beta}_{gj})\Phi^{-1}\left(1 - \frac{p_{gj}}{2}\right). \tag{17}$$

If m_g valid guides target the gene, their combined score and two-sided gene-level p-value are

$$Z_g = \frac{1}{\sqrt{m_g}} \sum_{j=1}^{m_g} z_{gj}, \qquad p_g = 2\Phi(-|Z_g|). \tag{18}$$

The reported gene effect is the median of the guide coefficient estimates,

$$\widehat{\beta}_g = \mathrm{median}_{j=1,\ldots,m_g} \widehat{\beta}_{gj}, \tag{19}$$

and gene-level false-discovery rates are obtained by applying Benjamini–Hochberg to the collection of p_g values. Consequently, concordant guide directions reinforce one another, discordant directions cancel, and one extreme guide has limited influence on the reported effect.

This aggregation is deliberately transparent, but it is not a native hierarchical gene model. Equations (17)–(18) give equal weight to valid guides and use the independence reference $\sqrt{m_g}$; shared sequence artifacts or other correlation can therefore make the gene statistic anti-conservative. A production extension should estimate guide reliability and within-gene dependence or use a hierarchical effect model. In the biological head-to-head benchmarks, this Stouffer–median procedure is used only for BARCS (and for the deliberately naive binomial comparator). MAGeCK, Chronos, BAGEL2, Waterbear, and MAUDE retain their native or deposited gene-level summaries. In the CRISPulator simulation sensitivity analysis, the same transparent rule is additionally applied to edgeR-QL, DESeq2, and limma-voom guide tests, as stated below, to make the absence of a native CRISPR gene model explicit. The exploratory DeWeirdt false-discovery audit used Fisher aggregation as stated in its Results description and is not pooled with these primary benchmark statistics.

Exchangeable normal guide-beta statistic. BARCS-NORM implements a different and deliberately simple model. It does not transform guide p-values and does not use guide-specific standard errors. Instead, for the m_g valid guides targeting gene g, it assumes

$$\widehat{\beta}_{gj} \overset{\text{iid}}{\sim} N(\mu_g, \sigma_g^2). \tag{20}$$

The gene effect and between-guide standard deviation are estimated directly:

$$\widehat{\mu}_g = \frac{1}{m_g} \sum_{j=1}^{m_g} \widehat{\beta}_{gj}, \tag{21}$$

$$s_g^2 = \frac{1}{m_g - 1} \sum_{j=1}^{m_g} (\widehat{\beta}_{gj} - \widehat{\mu}_g)^2. \tag{22}$$

Testing $H_0 : \mu_g = 0$ gives

$$T_g = \frac{\widehat{\mu}_g}{s_g/\sqrt{m_g}} \sim t_{m_g - 1} \quad \text{under Equation (20) and } H_0. \tag{23}$$

Thus normally distributed guide betas lead to a Student t reference when their unknown σ_g is estimated from the same guides. The software also reports $2\Phi(-|T_g|)$ as a plug-in normal sensitivity value, but it is not the primary p-value because it treats the estimated spread as known. Multiple guides support this guide-consistency calculation; they do not become additional biological screen replicates.

Random-effects and moderated random-effects statistics. BARCS-RE (the implementation label `BARCS-partial`) retains each guide-specific regression standard error and models

$$\widehat{\beta}_{gj} \mid \beta_g, \tau_g^2 \sim N\left(\beta_g,\, \mathrm{SE}(\widehat{\beta}_{gj})^2 + \tau_g^2\right). \tag{24}$$

It estimates τ_g^2 by the nonnegative DerSimonian–Laird estimator and then pools guide effects with weights

$$w_{gj} = \left\{\mathrm{SE}(\widehat{\beta}_{gj})^2 + \widehat{\tau}_g^2\right\}^{-1}, \qquad \widehat{\beta}_g = \frac{\sum_j w_{gj} \widehat{\beta}_{gj}}{\sum_j w_{gj}}. \tag{25}$$

BARCS-RE-EB (implementation label `BARCS-EB`) stabilizes the noisy gene-specific heterogeneity estimate by shrinking it toward the pooled screen-wide excess-Q estimate τ_0^2:

$$\widetilde{\tau}_g^2 = \frac{(m_g - 1)\widehat{\tau}_g^2 + d_0\tau_0^2}{(m_g - 1) + d_0}, \qquad d_0 = 4. \tag{26}$$

Both random-effects statistics are calibrated against prespecified negative-control genes when at least ten are available and otherwise against a robust whole-screen empirical null. In contrast, BARCS-NORM obtains its spread and reference degrees of freedom entirely within each gene. These three alternatives and BARCS-ST reuse exactly the same fitted guide coefficients; only the guide-to-gene inference changes.

Guide-consistency statistic for the one-replicate boundary. When sample replication is insufficient, converting a guide statistic through a low-degree-of-freedom t distribution can erase useful ranking information. We therefore prespecified a separate exploratory gene statistic, BARCS-GC, for the $R = 1$ CRISPulator analysis. It uses the uncalibrated guide coefficient and model-based standard error to estimate one shared gene effect. With

$$w_{gj} = \mathrm{SE}(\widehat{\beta}_{gj})^{-2}, \tag{27}$$

the inverse-variance estimate, its standard error, and raw Wald statistic are

$$\widehat{\beta}_g = \frac{\sum_{j=1}^{m_g} w_{gj}\widehat{\beta}_{gj}}{\sum_{j=1}^{m_g} w_{gj}}, \qquad \mathrm{SE}(\widehat{\beta}_g) = \left(\sum_{j=1}^{m_g} w_{gj}\right)^{-1/2}, \qquad T_g = \frac{\widehat{\beta}_g}{\mathrm{SE}(\widehat{\beta}_g)}. \tag{28}$$

This is a direct fixed-effect estimate of the common guide coefficient, not a Fisher or Stouffer combination of guide p-values. Its additional assumption is that the guide coefficients target one common gene effect; the reported direction-agreement fraction exposes gross heterogeneity.

Because T_g does not have a trustworthy sample-replicate reference distribution at $R = 1$, it is calibrated across genes. Let c_0 be the all-gene median and

$$s_{\mathrm{MAD}} = \frac{\mathrm{median}_g\,|T_g - c_0|}{\Phi^{-1}(0.75)}. \tag{29}$$

When at least ten valid control genes are present, let c_C be their median and let

$$s_C = \frac{Q_{0.95}\{|T_g - c_C| : g \in C\}}{\Phi^{-1}(0.975)}. \tag{30}$$

The calibrated statistic and two-sided empirical-null p-value are

$$Z_g^{\mathrm{GC}} = \frac{T_g - c_C}{\max(1, s_{\mathrm{MAD}}, s_C)}, \qquad p_g^{\mathrm{GC}} = 2\Phi(-|Z_g^{\mathrm{GC}}|), \tag{31}$$

followed by Benjamini–Hochberg adjustment. If too few control genes are available, c_0 replaces c_C and s_C is omitted. At least three finite guide coefficient estimates are required per gene. The all-gene MAD assumes that fewer than half of tested genes are active.

BARCS-GC measures reproducibility across independently designed perturbations; it does not turn guides into independent biological samples. Its p-value is consequently labeled empirical-null guide-consistency evidence, not sample-level confirmation. The ordinary BARCS result remains the primary analysis when biological replicates are available, and independent experimental replication remains necessary for confirmatory claims.

3.4.2 CRISPulator FACS benchmark

We used CRISPulator 0.5.1's documented interfaces for library construction, transfection, FACS selection, and sequencing [8]. The committed Julia environment pins the tagged source revision. For each simulation seed, one CRISPRn guide library was constructed and reused across four independently seeded transfection, sorting, and sequencing replicates. The default CRISPulator phenotype mixture was retained: 75% inactive, 5% negative-control, 10% phenotype-increasing, and 10% phenotype-decreasing genes. The headline screens contained 10,000 genes and five guides per gene, so 50,000 guides, which is the order of a genome-wide library rather than a pilot. The default guide-quality mixture—90% complete-knockout guides and 10% low-activity guides—was retained. Transfection representation was 500 cells per guide, Gaussian phenotype noise was 2, sorting representation was 50 cells per guide, and sequencing depth was 50 reads per guide per sample. Multiplicity of infection was 0.20 for the main analysis and 0.30 for the supplementary analysis; 0.20 is the point at which the single-integration approximation these guide-level models share is most defensible, and 0.30 stresses it. Both multiplicity of infection and the high-quality-guide fraction are exposed as simulation parameters, as is the number of genes and the number of independent screen replicates. The three genome-scale benchmark seeds were 20250724 through 20250726; the supporting 400-gene analyses below use 20250724 through 20250728.

Two BARCS gene statistics are carried through the genome-scale benchmark. BARCS-ST is the historical calibrated signed-z statistic. BARCS-MOD reuses the identical guide fits after guide-dispersion moderation, so the two differ only in how the guide variance is estimated, not in the gene combiner. The exchangeable-normal, random-effects, and moderated random-effects statistics are retained for the real-data comparisons but are not carried through the genome-scale simulation, where the question is whether moderation changes the calibration–power tradeoff rather than which of several pooling rules is best. Official MAGeCK-MLE 0.5.9.5 is included, with its ordered marker affinely mapped onto zero–one so that the low samples of the reference replicate form the all-zero reference row the MAGeCK initializer requires. Because MAGeCK-MLE omits negative-control genes from its gene summary once their sgRNAs are declared, all methods in the genome-scale benchmark are scored on the genes every method returned a finite result for, 9,475–9,517 of 10,000 per run. The negative-control diagnostic is instead taken from each method's own output, and is therefore unavailable for MAGeCK-MLE.

We additionally performed a one-factor-at-a-time sensitivity analysis around the baseline $(\mathrm{MOI}, q, G, R) = (0.25, 0.90, 400, 4)$, where q is the high-quality-guide fraction, G is the number of genes, and R is the number of independent screen replicates. The evaluated levels were $\mathrm{MOI} \in \{0.10, 0.25, 0.40\}$, $q \in \{0.60, 0.75, 0.90\}$, $G \in \{100, 400, 1000\}$, and $R \in \{1, 3, 4, 6\}$. After removing duplicate baseline settings, this yielded ten scenarios; each used the same five prespecified seeds, for 50 simulations. One parameter changed at a time, so its main sensitivity remained interpretable. For each seed and scenario, we paired BARCS and MAGeCK-MLE results from the low–bulk–high design and reported differences as BARCS minus MAGeCK-MLE. The executable benchmark also supports the complete $3 \times 3 \times 3 \times 4$ factorial (108 scenarios and 540 simulations across five seeds) when interaction effects are the target.

The $R = 1$ setting was prespecified as a diagnostic boundary rather than a confirmatory design. Its three low–bulk–high samples and two regression coefficients leave only one residual degree of freedom. The corresponding two-tail fit has two samples and two coefficients, leaves zero residual degrees of freedom, and was therefore not fitted. All pairwise-design comparisons and the primary performance interpretation use $R \geq 3$; the one-replicate result is shown separately to expose the consequence of insufficient replication. BARCS-GC was evaluated only at $R = 1$ using Equations (28)–(31); it was not used to replace ordinary BARCS in supported replicated designs.

CRISPulator's arbitrary percentile ranges were set to $(0, 0.25)$ for low, $(0.75, 1)$ for high, and $(0, 1)$ for bulk. Thus bulk deliberately overlaps the tails and is not a third component of a multinomial partition. An input sample was generated as an independent sequencing draw from the same guide frequency vector as bulk. Low and high received phenotype scores equal to the means of their corresponding standard-normal intervals,

$$d_{\text{low}} = -1.271, \qquad d_{\text{high}} = 1.271,$$

while bulk received $d_{\text{bulk}} = 0$. BARCS, MAGeCK-MLE, edgeR quasi-likelihood, DESeq2 Wald, and limma-voom fitted this score with replicate indicators [11, 24–26]. Tail-only fits removed bulk without changing the low and high counts. BARCS guide tests were scaled with the simulated negative-control guides before directional Stouffer aggregation; MAGeCK-MLE used its native gene-level Wald result. edgeR-QL, DESeq2, and limma-voom are general count-regression comparators rather than CRISPR gene models. Their native guide-level two-sided p-values and coefficients were therefore passed through the same unscaled directional Stouffer and median-effect summaries. This explicit hybrid comparison tests whether a strong general count model plus a common guide-to-gene rule is sufficient; it does not relabel that rule as a native gene statistic of those packages. Reported runtimes cover the primary low–bulk–high model fit and exclude simulation, file input/output, and guide-to-gene aggregation.

Genes from the increasing and decreasing classes were positives. Ranking used the two-sided gene significance score. AUROC and average precision measure active-gene discrimination. Effect recovery is Spearman correlation between the inferred gene effect and CRISPulator's median theoretical guide phenotype among active genes. Directional recall is the fraction of active genes with gene FDR below 0.10 and an effect sign matching the simulated class. Realized FDP is the fraction of called genes that were inactive, and F1 uses active versus inactive status at the same FDR threshold.

3.4.3 Liang Cas13 fitness benchmark

We used the transcriptome-scale RfxCas13d fitness screens of Liang et al. across HAP1, HEK293FT, K562, MDA-MB-231, and THP1 cells [5]. The library contains 56,322 guides targeting 5,496 lncRNAs, 3,156 protein-coding genes, and 1,000 non-targeting controls. The primary comparison used day 14 versus day 0 and the deposited normalized guide values in Table S2, retaining two biological replicates where available. Table S2C contains one K562 day-0 column and both day-14 columns. We used this deposited layout directly and applied no additional MAGeCK normalization. These are fractional processed values after the authors' median-of-ratios normalization, ComBat correction, and replicate-outlier procedure, not raw integer read counts. Because BARCS enforces integer counts, we explicitly rounded each value to the nearest pseudo-count, recorded the rounding error, and supplied the identical rounded matrix to BARCS, MAGeCK-RRA, and MAGeCK-MLE. Thus this analysis is explicitly a processed-count sensitivity comparison: BARCS's beta-binomial and MAGeCK's negative-binomial likelihoods do not retain their literal sampling interpretations. Its purpose is to compare rankings obtained from exactly the same public input, not to present pseudo-count p-values as a raw-count validation.

For an optional raw-count confirmation, the repository also supplies a restartable pipeline that streams endpoint reads from BioProject PRJNA1344834 without retaining FASTQ files. It implements the two stated Cutadapt anchor rules and aligns 23-nucleotide inserts to the Table S1B reference using Bowtie arguments `-v 1 -m 3 -best -q`. The article prose describes "up to three mismatches," whereas `-v 1` permits one; the executable argument is therefore the reproducible contract.

Four gene-level analyses were kept distinct. "Liang RRA" denotes the authors' deposited day-14 results from `RobustRankAggreg` 1.2.1 [27]; it is not MAGeCK-RRA. Official MAGeCK-RRA used `mageck test` without renormalizing Table S2, and official MAGeCK-MLE used a day-14 coefficient with a replicate indicator when two complete pairs were available [11, 15]. BARCS used the identical replicate-plus-day design, calibrated guide statistics with the 1,000 non-targeting controls, and applied the prespecified directional Stouffer gene summary. Each method retained its native gene statistic; agreement was reported separately as Spearman correlation of gene effects.

The published Liang RRA calls were treated as a comparator, not ground truth. The positive set comprised the library's known-essential protein-coding genes, defined in the study from DepMap knockout screens. Cell-line-specific nulls were targeted lncRNAs with TPM equal to zero in both available expression assays. Primary operating metrics were average precision, essential-control recall at an empirical 5% null false-positive rate, the absolute difference between the null $p < 0.05$ fraction and its nominal 0.05 target, and directional essential-control recall at gene FDR 0.10. Liang and BARCS gene p-values were Benjamini–Hochberg adjusted within each cell line; the official MAGeCK outputs supplied their gene FDR values. These controls test ranking and calibration without assuming that disagreement with the published discovery list is an error.

3.4.4 Legacy representation and local-equivalence result

We now separate an algebraic compatibility result from the substantive estimating-equation connection. The first result reproduces CB^2 after its weighted group summaries have already been computed; because a saturated two-cell GLS can represent any pair of independent group summaries, this result should not be interpreted as proof that the implemented `bbreg()` estimator strictly nests the original estimator. For the original test, let

$$\widehat{p}_A = \sum_{i \in A} a_i Y_i, \qquad \widehat{p}_B = \sum_{i \in B} b_i Y_i, \qquad \sum_{i \in A} a_i = \sum_{i \in B} b_i = 1,$$

where the depth- and dispersion-dependent weights a_i, b_i are those estimated by CB^2. Let its estimated group variances be $V_A > 0$ and $V_B > 0$. The original guide statistic and reference degrees of freedom are

$$T_{\mathrm{CB2}} = \frac{\widehat{p}_B - \widehat{p}_A}{\sqrt{V_A + V_B}}, \tag{32}$$

$$\nu_{\mathrm{CB2}} = \frac{(V_A + V_B)^2}{V_A^2/(n_A - 1) + V_B^2/(n_B - 1)}. \tag{33}$$

Lemma 1 (Legacy CB^2 representation)**.** *Let X have row $(1, 0)$ for samples in group A and row $(1, 1)$ for samples in group B. Define the working precision matrix*

$$\Omega = \operatorname{diag}\left(\left\{ \frac{a_i}{V_A} : i \in A \right\}, \left\{ \frac{b_i}{V_B} : i \in B \right\} \right).$$

Then the identity-link generalized least-squares estimator satisfies

$$\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (X^\top \Omega X)^{-1} X^\top \Omega \mathbf{Y} = \begin{pmatrix} \widehat{p}_A \\ \widehat{p}_B - \widehat{p}_A \end{pmatrix},$$

and

$$(X^\top \Omega X)^{-1} = \begin{pmatrix} V_A & -V_A \\ -V_A & V_A + V_B \end{pmatrix}.$$

Consequently, the GLS t statistic for the slope is exactly T_{CB2}.

Proof. Because the weights within each group sum to one, their groupwise total working precisions are

$$\sum_{i\in A}\frac{a_i}{V_A}=\frac{1}{V_A},\qquad \sum_{i\in B}\frac{b_i}{V_B}=\frac{1}{V_B}.$$

Hence

$$X^\top\Omega X=\begin{pmatrix}V_A^{-1}+V_B^{-1} & V_B^{-1}\\ V_B^{-1} & V_B^{-1}\end{pmatrix},\quad X^\top\Omega\mathbf{Y}=\begin{pmatrix}\widehat{p}_A/V_A+\widehat{p}_B/V_B\\ \widehat{p}_B/V_B\end{pmatrix}.$$

Direct inversion gives the displayed covariance matrix and multiplication gives the displayed coefficient vector. Its slope standard error is $\sqrt{V_A+V_B}$, proving the result. □

Corollary 1 (Legacy-compatibility identity)**.** *If the slope statistic in Lemma 1 is referred to $t_{\nu_{\mathrm{CB2}}}$ using Equation* (33)*, its two-sided p-value is identical to the original CB^2 p-value. Thus identity-link GLS algebra reproduces the original two-group inference on its group-summary scale. This corollary is a compatibility check, not a claim of finite-sample equality with the default logit `bbreg()` fit.*

The default implementation uses the logit link because it respects the parameter space for arbitrary designs. It is therefore important to prove a second, weaker relationship rather than claim finite-sample identity.

Proposition 1 (First-order equivalence of the logit contrast)**.** *Suppose the two groups are independent, their weighted proportion estimators are consistent, their groupwise and pooled precision estimators converge to a common κ, and under the null or a sequence of local alternatives $p_A, p_B \to p \in (0,1)$ with $\widehat{p}_B-\widehat{p}_A=O_p(n^{-1/2})$. Let $h(p)=\mathrm{logit}(p)$ and define the delta-method statistic*

$$T_{\mathrm{logit}}=\frac{h(\widehat{p}_B)-h(\widehat{p}_A)}{\sqrt{h'(\widehat{p}_A)^2V_A+h'(\widehat{p}_B)^2V_B}}.$$

Then

$$T_{\mathrm{logit}}-T_{\mathrm{CB2}}\xrightarrow{p}0.$$

Under a correctly specified common-dispersion beta-binomial model, the binary-design IRLS Wald statistic has the same first-order expansion. If $n_A, n_B\to\infty$ through increasing numbers of independent biological libraries, both its residual degrees of freedom and Equation (33) *diverge, so their reference distributions also converge to the same standard normal limit. Holding n_A, n_B fixed while only increasing read depths N_i is not this asymptotic regime.*

Proof. The mean-value theorem and consistency give

$$h(\widehat{p}_B)-h(\widehat{p}_A)=h'(p)(\widehat{p}_B-\widehat{p}_A)+o_p(n^{-1/2}),$$

where $h'(p)=\{p(1-p)\}^{-1}$. Continuity of h' gives

$$\sqrt{h'(\widehat{p}_A)^2V_A+h'(\widehat{p}_B)^2V_B}=h'(p)\sqrt{V_A+V_B}\{1+o_p(1)\}.$$

The common positive factor $h'(p)$ cancels, yielding $T_{\mathrm{logit}}=T_{\mathrm{CB2}}+o_p(1)$. It remains to connect this transformed cell-mean statistic to IRLS. Write $\rho=1/(\kappa+1)$. Within a group having mean μ_g, the beta-binomial quasi-score for its logit cell mean is proportional to

$$\sum_{i\in g}\frac{N_i}{\kappa+N_i}(Y_i-\mu_g).$$

The original CB^2 weight is

$$\left(\frac{1}{\kappa}+\frac{1}{N_i}\right)^{-1}=\frac{\kappa N_i}{\kappa+N_i}.$$

The factor κ disappears on normalization, so for fixed common dispersion the two methods use identical within-group relative weights. Under the stated common-dispersion model, the separately estimated legacy precisions and the IRLS precision converge to the same κ; the resulting binary-design cell means therefore have the same first-order expansion. Finally, both t distributions approach the standard normal as their degrees of freedom diverge. $\square$

These results establish two deliberately separate claims. The original statistic has an *algebraically exact legacy representation* after its group summaries are computed, while the implemented logit statistic is *locally asymptotically equivalent* in the unadjusted binary design under the stated common-dispersion assumptions. Strict finite-sample nesting of the implemented estimator is not claimed. In finite samples, `bbreg(~group)` need not equal `measure_sgrna_stats()`: the effect scale, dispersion pooling, working weights, and degrees of freedom differ. What regression adds is an estimable coefficient for continuous predictors, adjustment variables, and interactions; what it changes must be assessed by calibration and benchmark data rather than hidden behind the word "generalization." The executable example `examples/cb2_generalization_proof.R` verifies Lemma 1 to machine precision and illustrates the convergence in Proposition 1. In its deterministic guide, original CB^2 and the GLS contrast both give effect 9.059929×10^{-4}, standard error 1.113105×10^{-4}, $t=8.139330$, $\nu=5.647255$, and two-sided $p=2.512928\times10^{-4}$; the maximum numerical discrepancy is 1.78×10^{-15}. In the link-scale illustration, shrinking the two-group proportion difference from 8×10^{-4} to 2.5×10^{-5} reduces the absolute gap between raw and logit statistics from 0.0583 to 0.000151.

4 Software and usage

The standalone beta-binomial implementation uses base R. The same additive API is integrated into the cloned CB^2 package; there its weighted IRLS cross-products and solve are implemented with RcppArmadillo. Existing CB^2 two-group functions remain unchanged: BARCS is an additive analysis path rather than a breaking replacement.

```
source("R/bbreg.R")

# One guide: K contains guide counts; N contains sample library sizes.
fit <- bbreg(
  count   = K,
  total   = N,
  formula = ~ scale(dose) + batch,
  data    = sample_data
)
summary(fit)

# A named contrast; unspecified coefficient weights are zero.
bb_contrast(fit, c("scale(dose)" = 1))

# A complete guide-by-sample screen.
result <- bb_screen(
  counts  = count_matrix,
```

```
  totals  = N,
  data    = sample_data,
  formula = ~ scale(dose) + batch,
  term    = "scale(dose)",
  gene    = guide_annotation$gene,
  ncores  = 4
)

# Optional empirical-null scaling when prespecified negative controls exist.
result <- bb_calibrate_controls(
  result,
  control = guide_annotation$gene == "Non-Targeting Control"
)
result[order(result$fdr), ]

# Optional direct shared-effect gene inference; no Fisher/Stouffer step.
gene_result <- bb_gene_consistency(
  result,
  control = guide_annotation$gene == "Non-Targeting Control"
)
```

Column order in the count matrix must match row order in `sample_data`. The phenotype should be defined before looking at guide results, and all independent biological replicates should remain separate. For the negative-binomial benchmark, the project calls the official `mageck mle` executable with the same design matrix rather than embedding a partial MAGeCK reimplementation in CB^2. On the development machine, the installed CB^2 package analyzes 10,000 guides in 4.73 seconds serially and 1.36 seconds with four forked workers. The longitudinal analysis fits 86,882 filtered guides across eight aggregated time points in 19.4 seconds with four workers (4,481 guides per second); 86,840 fits reached the strict convergence tolerance, and all fits returned finite time coefficients.

5 Discussion: the problem BARCS is designed to solve

The methods answer related but different questions. Original CB^2 remains the direct choice when the biological estimand is one prespecified difference between two independent conditions. BARCS should be preferred when the design itself carries information that a pairwise split would discard: three or more ordered levels, a dose or time axis, donor or batch adjustment, an interaction, or a named contrast from a multivariable model.

This decision rule defines the problem BARCS is intended to solve. It does not ask users to abandon a validated two-group workflow or replace a purpose-built generative model. It supplies the missing middle layer: a depth-aware coefficient test when the observed units are independent libraries and the scientific question is a trend, adjusted association, interaction, or contrast rather than one group label.

6 Assumptions, diagnostics, and limitations

1. **Independent samples.** The t_{m-q} reference distribution assumes independent sequenced libraries. Repeated measures, matched samples, or multiple libraries from one biological unit

Table 15: Practical method choice by experimental design.

Design	Primary method	Reason
Two independent groups	■ CB^2 or ■ MAGeCK	The estimand is already a pairwise contrast
Independent libraries across dose, time, or ordered phenotype	■ BARCS	Uses the full quantitative axis with sample-level t inference
Multivariable independent samples	■ BARCS	Adjusts batch, donor, and interactions in one explicit model matrix
FACS bins partitioning the same donor pool	■ Waterbear	Models the joint bin probabilities and within-donor dependence
Multi-line viability dynamics	■ Chronos, with ■ BARCS as a transparent trend analysis	Mechanistic growth and knockout timing may matter
Cancer screen with suspected CNV artifact	Diagnose before correction	Post-fit CNV normalization can help, do nothing, or over-correct

require a subject-level correlation model or a blocked permutation. FACS bins from one donor are a compositional partition, not four independent biological replicates; the GSE242880 control calibration is a diagnostic sensitivity analysis, whereas a purpose-built joint model such as Waterbear represents that structure directly. Likewise, a 0–100% bulk pool overlaps its sorted tails. The CRISPulator analysis includes that requested design to measure its empirical behavior, not to redefine bulk as an independent third bin.

2. **Enough sample-level degrees of freedom.** Read depth cannot replace biological replication. With very few samples, guide-wise $\widehat{\rho}$ is noisy and coefficient tests may be poorly calibrated. The model-based covariance treats $\widehat{\rho}$ as a fixed plug-in value, and the t_{m-q} reference does not formally propagate its estimation uncertainty. The local-equivalence result likewise requires increasing numbers of independent libraries; deeper sequencing at fixed replication is not sufficient. The optional `bb_calibrate_controls()` procedure additionally requires all control guides to share one residual degree of freedom, as they do when every guide uses the same complete design. Guides fitted with heterogeneous designs require stratified or redesigned calibration. The optional `bb_gene_consistency()` statistic can recover concordant multi-guide signal when sample-level degrees of freedom are inadequate, but its empirical-null p-values measure guide consistency. They do not repair the missing biological replication and should not be described as confirmatory sample-level inference. Dispersion shrinkage across guides or permutation of the phenotype should be evaluated before treating this prototype as a production method.

3. **Mean-model form.** A single slope assumes a linear relationship on the logit scale. Residual plots should be checked. A spline or a prespecified nonlinear term is preferable when the dose response is curved. This matters directly in viability time courses: incomplete knockout can produce finite depletion saturation, which Chronos models mechanistically but a single beta-binomial slope does not.

4. **Sparse and separated guides.** Guides with all zero counts or with counts confined to one end of the phenotype range can cause boundary fits. Low-count filtering should be chosen without reference to the tested phenotype. A penalized or likelihood-ratio procedure is a useful future extension for severe separation.

5. **Compositional counts.** Guide counts share a fixed library total. The marginal beta-binomial

model handles depth and guide-specific overdispersion but does not remove global composition shifts. Stable nontargeting controls and sensitivity analyses with alternative size factors are advisable. If guides are filtered for testing or benchmarking, the `totals` argument must retain the unfiltered mapped-read totals. Recomputing totals after row subsetting changes the likelihood and can materially change calibration.

6. **Guide-to-gene inference.** The BARCS regression itself is guide-level and requires no Fisher or Stouffer combination. Some replicated benchmark analyses use a directional Stouffer summary only as a transparent, explicitly labeled comparison adapter. BARCS-GC instead estimates one shared guide coefficient by inverse-variance weighting and calibrates its Wald statistic against a robust gene-level empirical null for the one-replicate diagnostic. That fixed-effect construction assumes a common gene effect, while its all-gene MAD assumes that fewer than half of genes are active. The direction-agreement output should be inspected for guide heterogeneity. We additionally evaluated BARCS-NORM, which treats the fitted guide coefficients within a gene as exchangeable draws $\widehat{\beta}_{gj} \sim N(\mu_g, \sigma_g^2)$ and tests the sample mean with a t_{m_g-1} reference because σ_g is estimated. This literal normal model was calibrated but underpowered with three to six guides: across the 50 supporting 400-gene CRISPulator runs its average precision was 0.706 and its directional recall at FDR 0.10 was 0.159. Replacing the t reference by a plug-in normal tail would incorrectly treat the estimated σ_g as known. A production hierarchical gene test should therefore borrow variance information across genes and be benchmarked separately, ideally with held-out controls or phenotype permutations that preserve cross-guide dependence.

7. **Comparator assumptions.** MAGeCK-MLE offers information sharing and guide-efficiency estimation that the prototype lacks, but its normalized negative-binomial model and asymptotic Wald inference must also be diagnosed. Head-to-head results should be stratified by replication, phenotype type, signal strength, and composition shift rather than treating either method as a universal reference. The Liang benchmark illustrates the converse limitation for BARCS: with one residual degree of freedom and preprocessed pseudo-counts, the study-specific rank aggregation is more powerful and better calibrated than either regression likelihood. Likewise, post-fit CNV correction should be accepted only after a null-gene diagnostic: it succeeds in A375 and over-corrects the longitudinal HT-29 screen.

7 Conclusion

BARCS solves one defined problem: testing a prespecified guide-abundance coefficient when independent pooled-screen units lie on a quantitative or covariate-adjusted design. It preserves full-library denominators, separates read-depth precision from extra-binomial heterogeneity, and assigns reference degrees of freedom to independent libraries. The scientific output can therefore be a time slope, dose response, adjusted treatment effect, interaction, or named contrast without reducing the experiment to an arbitrary binary split.

The benchmarks support that focused claim. BARCS is competitive with continuous-time MAGeCK and Chronos in longitudinal HT-29 data, provides the strongest high-precision recall there, and recovers 22 of 26 validated IL2RA regulators after control calibration in the ordered-bin stress test. The genome-scale CRISPulator study shows its characteristic tradeoff: no universal advantage in global ranking, where the six methods compared span 0.019 in average precision, but a requested error rate that is close to the realized one, which the count models tested here do not deliver. Endpoint analyses recover established biology and demonstrate backward compatibility rather than a need to replace successful pairwise tools.

Equally important, the method has explicit stopping rules. It does not create biological replication from read depth or guide number, make bins from one donor independent, infer mechanistic growth dynamics, or guarantee that post-fit copy-number correction helps. Liang's processed Cas13 benchmark shows the loss of threshold power with one residual degree of freedom; Waterbear and Chronos remain preferable when joint-bin dependence or mechanistic viability is the estimand. BARCS-GC can prioritize a conservative multi-guide subset in an otherwise irreplaceable one-replicate screen, but its empirical-null Wald evidence remains exploratory.

The contribution is therefore a bridge, not a universal replacement. Original CB^2 answers a two-condition question; BARCS answers a coefficient question using the same depth-aware beta-binomial principle. The next steps follow directly from the remaining gaps: empirical-Bayes dispersion moderation, a hierarchical gene-effect model, blocked or random-effect inference for repeated biological units, nonlinear dose and time effects, and covariance estimators that propagate dispersion uncertainty.

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